How Many Atoms, Where, and When: Predicting the Discrete Optima of Peak-Power Capacity, Minimax Estimation and Nonparametric Maximum Likelihood Under Gaussian Noise

I Made Tristan Hope Firdaus
Independent Researcher, Bali, Indonesia
E-mail: madetristanfirdaus@gmail.com ORCID: 0009-0002-3048-3624

***Abstract*—We study three discrete optimal laws under Gaussian noise: the capacity-achieving input of a peak-limited channel, the least favorable prior for a bounded mean, and the nonparametric maximum-likelihood estimate of a mixing law. We ask how many atoms they have, where they sit and when a new one appears as the region grows. On an interval of half-length A we prove that the capacity-achieving input has at least $0.1660\,A^{4/3}$ atoms for large A. This improves the best published lower bound, of order $A\sqrt{\log A}$, and has the order of the numerical law $0.4002\,L^{4/3}$ of all three problems, where L is the half-length widened by a fixed edge strip. The atoms move outward as the amplitude grows if one matrix is totally positive, which we prove for its Gaussian part and, by a computer-assisted proof, at 91 optima. The amplitudes at which a center atom is born or splits follow a series law that predicted three blocks of eight later events inside bands fixed in advance. On a closed curve near a circle we conjecture, and test on events predicted in advance, that the atoms are the nodes of a Szegő quadrature of a signed density computed from the optimality condition. A computer-assisted proof shows that, for one ellipse, the capacity-achieving input of the two-by-two peak-limited multiple-input multiple-output channel has no atoms at the ends of its major axis, which contradicts a published theorem. The other laws stated here are measured, and their tests are numerical.**

***Index Terms*—Amplitude-constrained Gaussian channel, capacity-achieving distribution, discrete optimal inputs, least favorable prior, nonparametric maximum likelihood, total positivity.**

I. Introduction

Consider a transmitter limited in peak power, a statistician who knows that an unknown mean lies in a bounded region, and a quantizer for a source spread over a region. All three face the same mathematical situation: a bounded quantity is seen through additive Gaussian noise, and a probability law over that quantity has to be chosen optimally. For the transmitter the law is the input distribution that maximizes the mutual information. For the statistician it is the least favorable prior (LFP) of squared-error estimation. For the quantizer with squared-error distortion, the optimal reproduction law is found by fitting a Gaussian mixture to the source by maximum likelihood [1]; for a source spread uniformly, this fit is the population nonparametric maximum-likelihood estimate (NPMLE) of the mixing law. In all three problems the optimal law is discrete. Smith proved this for the scalar channel [2]. For the bounded normal mean it follows from Ghosh's work [3] (see [4] and [5, Prop. 4.18]), and Polyanskiy and Wu bounded the number of atoms of the NPMLE of a one-dimensional sample by a constant times the squared range of the data [6, Thm. 1]. Polyanskiy and Sellke compute that estimator to any accuracy and certify its exact number of atoms in finite time [7]. On a disc or a ball the capacity-achieving input sits on finitely many concentric circles or spheres [8]–[10].

How many atoms these laws have, where the atoms sit, and when a new atom appears as the region grows are known in much less detail. On an interval of half-length A the capacity-achieving input has between order $A\sqrt{\log A}$ and order A^2 atoms by the published bounds [11], [12], and about $A^{4/3}$ according to computations and a non-rigorous derivation [13], [14]; Wang [15] gave the first super-linear lower bound. Barletta and Dytso [16] show that an input within ϵ of capacity needs order $A\sqrt{\log A}$ atoms when ϵ decays polynomially in A, and they suggest that the different exponents seen in numerical studies belong to different regimes of ϵ and not to the exact optimizer. Every local change of the capacity-achieving support happens at the origin, where an atom appears or disappears, or the atom there splits into a pair or a pair merges [17]. In two dimensions even less is known. Dytso, Barletta and Kramer reduced the 2×2 multiple-input multiple-output (MIMO) channel with a peak-power constraint to an input confined to a filled ellipse [18]. They proved that for semi-axes at most $\sqrt{2}$ the optimal input lies on the boundary ellipse and, unless the ellipse is a circle, has finitely many atoms. They also stated that both ends of the major axis always carry an atom, and they left open how the number of atoms grows as the ellipse becomes a circle. An earlier condition of Dytso, Goldenbaum, Poor and Shamai, obtained from the maximum principle for subharmonic functions, puts the support on the boundary of any input region whose image under the channel matrix lies in a small enough ball [19, Thm. 8].

The first result of this paper concerns the exact optimizer on an interval.

Theorem 1. *Let $K(A)$ be the number of atoms of the capacity-achieving input of $Y = X+Z$ with $|X| \le A$ and $Z \sim \mathcal{N}(0,1)$. Then $\liminf_{A\to\infty} K(A)/A^{4/3} \ge \frac{1}{4}(4\rho_1)^{-1/6} > 0.1660$, where $\rho_1 = (e-1)\lambda_1/(1-e^{-\lambda_1})$ and $\lambda_1 = 1.1648$.*

The proof is in Appendix A. It uses the Kuhn–Tucker conditions, the structure of the Gaussian kernel and one evaluation in interval arithmetic. It raises the order of the best published lower bound from $A\sqrt{\log A}$ to $A^{4/3}$, so the exact optimizer has at least the order of the numerical law of Section III; whether it has more is open, because only A^2 is known from above.

The other results are the following.

- *A law for births on a line.* The amplitudes at which the center atom is born or splits follow the births law, whose form has no free parameter. The order of its series, the spline and the depth of its half-line problem (the same problem with a single wall) were chosen on known events (Section IV-B). It reproduces 20 capacity events to within 0.09×10^{-9}. Three blocks of eight later events, predicted before they were computed, fell inside acceptance bands fixed in advance.
- *Outward motion and monotonicity.* That the atoms move outward as the amplitude grows follows if one matrix is totally positive. We prove this for its Gaussian part (Theorem 2) and, by a computer-assisted proof, the whole matrix at 91 optimal states (Proposition 1). The monotonicity of the number of atoms reduces to two statements about the innermost atoms. They hold in every state we stored, they are proved at the optimal states of Proposition 1, and they are not proved in general (Section V-A).
- *A rule for rings.* On a closed curve near a circle, we conjecture that the optimal ring of atoms is the Szegő quadrature of the formal density of Section II, which need not be positive. We also conjecture that the ring stops being optimal where a Verblunsky coefficient of that density reaches ± 1 (Conjectures 1 and 2). Of fourteen capacity events registered in advance, the ten on the two curves closest to a circle fell within 0.28% of the rule alone, and the six for which a linear-response correction was computed fell within 10^{-4} (relative). That correction is a first-order shift of the predicted event (Section VI-C). Ten more, in the NPMLE, the LFP and deterministic annealing, fell within 0.61% of the rule alone, four of them within 3.5×10^{-6} with the correction (Section VI).
- *A counterexample, by a computer-assisted proof.* The rule predicted rings with no atom at the ends of the major axis. We prove one such ring optimal by a computer-assisted proof in interval arithmetic, so the vertex statement of [18] does not hold (Theorem 3, Section VII). The formal density of the rule also predicts where the optimum on a filled region first needs an atom inside it. Registered predictions of that point held to 3×10^{-7} on ellipses in all three problems and, for the NPMLE and the LFP, on mixed curves (Section VIII).

Theorems 1 and 2 and every proposition and lemma are proved in this paper; the proofs of Theorem 3 and Proposition 1 are computer-assisted proofs in interval arithmetic. The two-sided bounds on the capacity edge constant β in Section III are proved there, with two evaluations in interval arithmetic. Everything else is numerical and is labeled as a measured law or a conjecture where it appears. The registration log, the solvers, the computed states and the logs of the runs reported here form the supplementary material, at https://github.com/Tristan-tech-ai/how-many-atoms. The predictions were recorded in a dated log kept in that repository. From 14 October 2026 on, a hash of the log is time-stamped after each new entry by an independent time-stamp authority and anchored in the Bitcoin block chain through OpenTimestamps (folder `registration-log/timestamps`); the dates of earlier entries rest on the log itself. Supplementary Notes, submitted with this paper, collect the side results that do not carry the argument: optimal values and center events in d dimensions, the inner rings of the planar rule, and the residual of the births law.

Terms. A prediction is *registered* when its value and an acceptance band (its bar) were appended to a dated, append-only log before the optimum was computed or read, and *blind* when the optimum did not yet exist. A result is *certified* when a computer-assisted proof establishes it: every quantity is computed in ball arithmetic, the interval arithmetic in which each number is a midpoint with a radius that bounds all rounding and truncation errors (Section VII), so that each computed interval contains the exact value. A *measured law* is a relation found on computed optima and tested on later ones; it is not proved. The *filled region* of a curve is the region it bounds, and a *tracked optimum* is an optimum followed by continuation as a parameter changes. A *ring* is an optimum whose atoms all lie on the boundary curve; it *copies* the formal density when its low Fourier coefficients equal the density's. A *loss event* is the parameter value at which an optimum stops satisfying the optimality condition; the ring is then *lost*. At a *birth* a new atom appears and at a *split* one atom becomes two. A *stored state* is a computed optimum kept in the repository with its Kuhn–Tucker residual. The *edge strip* of a problem is the widening of an interval described by its one-dimensional edge constant (Section III), and the *linear-response correction* is the first-order shift of a loss event derived in Section VI-C. Following the convention of each literature, we write A, m and R for the half-width or radius in the capacity, LFP and NPMLE problems; r_p and r_m are the semi-axes of an ellipse. A list of the symbols of the paper, each with the section where it is defined, opens the Supplementary Notes.

What the results are for, and what they do not show. Each of the three optima is used, and each use needs its atoms. A transmitter with a peak limit sends a finite constellation, and the capacity-achieving input is the constellation that carries the most information. A minimax estimator for a mean known to lie in a region is the Bayes rule against the least favorable prior, which otherwise has to be found by a search over discrete priors. Algorithms for the NPMLE work on a grid, and deterministic annealing follows splits as a temperature falls; the count and the predicted birth points say how fine the grid must be and where the splits come. The certified ring corrects a published statement: the optimal input of the 2×2 peak-limited MIMO channel need not use the ends of the major axis, so a design or a proof that assumes those atoms needs another argument. We have not compared these optima with practical schemes, so we cannot say how much rate, risk or

likelihood a designer gains from them.

Section II states the three problems and their common condition. The results on an interval come first: the one-dimensional facts with the count law and the lower bound (Section III), births on a line (Section IV), and outward motion and monotonicity (Section V). The results on closed curves and filled regions follow: the rule for rings, the certified ring and the point where the optimum on a filled region first needs an atom inside it (Sections VI–VIII). Section IX lists what does not carry over and what is open. The appendices give the proofs, in the order of first use.

II. Three problems and one optimality condition

Let Z be a standard Gaussian vector in $\mathbb{R}^d$ with density φ_d, and let $\Omega \subset \mathbb{R}^d$ be a compact region. A probability law π on Ω is observed through the noise, $Y = X + Z$ with $X \sim \pi$, and its output density is $p_\pi = \pi * \varphi_d$. The three problems choose π as follows.

- *Capacity:* $C(\Omega) = \max_\pi I(X;Y)$, in nats.
- *LFP:* the Bayes risk of π for estimating X from Y under squared loss is $B(\pi) = \mathbb{E}|X - \mathbb{E}[X \mid Y]|^2$. The LFP maximizes B, and its Bayes risk is the minimax risk.
- *NPMLE:* for data with density f_{data}, the population NPMLE [20] maximizes $\int f_{\mathrm{data}} \log p_\pi$ over π. For data uniform on Ω, with density $f_{\mathrm{data}} = f_0 = 1/|\Omega|$ there, we report $G(\pi) = \int_\Omega \log(p_\pi/f_0)\, dy$.

Read with f_{data} as a source and squared error as the distortion, the NPMLE is the rate–distortion problem at the slope at which the test channel has unit variance [1]. In Section III the source is uniform on Ω; in Sections VI and VIII it is Gaussian, $N(0, s^2 I)$ with $s = 2$, and the reproduction law is confined to Ω.

Each objective is concave in π, so its maximum is characterized by its derivatives toward point masses. For a law π and a point q, let

$$D_\pi(q) = D\big(\varphi_d(\cdot - q) \,\big\|\, p_\pi\big) \quad \text{(capacity)}, \tag{1}$$

$$r_\pi(q) = \mathbb{E}\,|\delta_\pi(q + Z) - q|^2 \quad \text{(LFP)}, \tag{2}$$

$$J_\pi(q) = \int \varphi_d(y - q)\, \frac{f_{\mathrm{data}}(y)}{p_\pi(y)}\, dy \quad \text{(NPMLE)}, \tag{3}$$

where $\delta_\pi(y) = \mathbb{E}_\pi[X \mid Y = y]$ is the Bayes rule. Moving a small mass from π to q changes the objective at the rate $D_\pi(q) - I(\pi)$, $r_\pi(q) - B(\pi)$ or $J_\pi(q) - 1$. Hence π is optimal if and only if the function is at most a constant on Ω, with equality on the support of π. The constant is C, $B(\pi)$ or 1 (for capacity see [2], [21] and Lemma 1 of [18]; for the LFP, [4] and [5, Ch. 4]; for the NPMLE, [22]). We call this function the optimality function of the problem and write it F_π when the problem does not matter.

Two identities are used later. By Tweedie's formula [23], [24] the Bayes estimator is $\delta_\pi(y) = y + \nabla \log p_\pi(y)$. With $Y = X + Z$ its risk is therefore $B(\pi) = \mathbb{E}|Z + \nabla \log p_\pi(Y)|^2 = d + 2\,\mathbb{E}\langle Z, \nabla \log p_\pi(Y)\rangle + I(p_\pi)$, where $I(p_\pi) = \mathbb{E}|\nabla \log p_\pi(Y)|^2$ is the Fisher information of the output density. Stein's identity and an integration by parts give $\mathbb{E}\langle Z, \nabla \log p_\pi(Y)\rangle = \mathbb{E}\,\Delta \log p_\pi(Y) = -I(p_\pi)$, so $B(\pi) = d - I(p_\pi)$; we checked this on the computed LFP of the disc of radius $m = 2.5$ to 2×10^{-12}. For capacity, the Hessian of D_π is $\nabla^2 D_\pi(q) = I - \mathbb{E}_Z \operatorname{Cov}(X \mid q + Z)$ [25].

The atoms of an optimum are hard to predict one by one, but a smooth law is easier to compute. Let $\Gamma = \{z(t) : t \in [0, 2\pi)\}$ be a closed curve, and for a density g on Γ (with respect to $dt/2\pi$) let $F(t; g)$ be the optimality function at $z(t)$. The *formal density* of order M is the density $f(t) = 1 + \sum_{k=1}^{M} c_{2k} \cos 2kt$ whose optimality function has no harmonics of orders $2, \dots, 2M$, so that $F(\cdot\,; f)$ is constant along Γ to that order. We compute it by Newton's method in high-precision arithmetic. The optimality function of a signed f is defined wherever its output density is positive; for the truncations we use it was positive at every point where we evaluated it. Since the coefficients of f grow with k and f is negative on part of the curve from some order on, f is not an admissible law. We use only its low harmonics, which are what the optimal atoms follow.

III. One dimension: edge constants, the count and a lower bound

A. Edge constants and the count law

On an interval $[-A, A]$ the optimal atoms are spaced unevenly in all three problems: they lie farthest apart next to the two ends and come closer together toward the middle. Each problem therefore has an edge layer, and a constant describes it (Table I). For capacity, numerically, $e^{C+h_1} = 2A + \beta$ up to terms that fall off exponentially in A, where $h_d = \frac{d}{2}\log 2\pi e$ is the entropy of the noise; each end thus adds a strip of width $\beta/2 = 1.9908$ to the interval. For every $A > 0$ the test density of [26, eq. (12)] and the entropy-power inequality for a uniform input on $[-A, A]$ give $\sqrt{4A^2 + 2\pi e} \le e^{C+h_1} \le 2A + \sqrt{2\pi e}$, so that $C = \log 2A - h_1 + o(1)$ and a limit β of $e^{C+h_1} - 2A$, where it exists, lies in $[0, \sqrt{2\pi e}]$. An explicit input gives a sharper lower bound: Lebesgue measure on $[-A+g, A-g]$ plus atoms of masses ν_k at $\pm(A + s_k)$ with $-g \le s_k \le 0$, normalized by $M = 2A + 2(\sum_k \nu_k - g)$. If G_A is the convolution of this measure with φ, the output entropy is exactly $\log M - M^{-1}\int G_A \log G_A$. Near the right end $G_A(A + t)$ tends to $G(t) = \Phi(-g - t) + \sum_k \nu_k \varphi(t - s_k)$, the same holds at the left end, and $G_A \log G_A \to 0$ in the bulk, where $G_A \to 1$. Hence $\int G_A \log G_A = 2E + o(1)$ with $E = \int_{\mathbb{R}} G \log G$, and since $e^{C+h_1} \ge e^{h(X+Z)}$ for every admissible input, $\liminf_{A\to\infty}(e^{C+h_1} - 2A) \ge 2(\sum_k \nu_k - g - E)$. For $g = 4.361045$ and $(s_k, \nu_k) = (0, 3.49413)$, $(-2.68983, 2.25076)$, $(-4.36104, 0.62400)$, the integral evaluated in ball arithmetic, with the tails bounded in closed form, gives $2(\sum_k \nu_k - g - E) > 3.97966$; the measured β lies less than 0.002 above this proved bound. A profile $\tilde{G}$ of the same form, with five atoms and the innermost at $s_5 = -g$, gives an upper bound; it serves only as a test density, so its atoms need not lie in $[-g, 0]$. We take $g = 4.218943865$ and $(s_k, \nu_k) = (0.014712017, 3.435818087)$, $(-2.423658261, 1.584446936)$, $(-3.337072961, 0.131437977)$, $(-3.242557666, 0.645464019)$, $(-4.218943865, 0.416780629)$, and use the test density

$q = \tilde G_A/\tilde M_A$ with $\tilde G_A(y) = \tilde G(y-A)\,\tilde G(-y-A)$ and $\tilde M_A = \int \tilde G_A$. Then $D(P_{Y|x}\|q) = \log \tilde M_A - h_1 - \vartheta(x-A) - \vartheta(-x-A)$ with $\vartheta(u) = \int \varphi(t-u)\log \tilde G(t)\,dt$, and $\vartheta \ge 0$ on $(-\infty, 0]$, by the second certificate described below. Since $C \le \sup_{|x|\le A} D(P_{Y|x}\|q)$ [26], $e^{C+h_1} \le \tilde M_A$ for every $A > 0$, and $\tilde M_A - 2A \to 2\int(\tilde G - \mathbf{1}_{(-\infty,0)}) = 2(\sum_k \nu_k - g)$ by dominated convergence. Hence $3.97966 < \liminf_{A\to\infty}(e^{C+h_1} - 2A) \le \limsup_{A\to\infty}(e^{C+h_1} - 2A) \le 3.99001$.

The two certificates. Both are computed in ball arithmetic at 160 bits, with the library of Section VII-B. For the lower bound, E is integrated on $[-40, 40]$ by its certified integrator, with the interval split at $-g$ and at the atoms. The tails are bounded in closed form, by $|G\log G| \le \sqrt{G}$ for $t \ge 40$ and by $|G\log G| \le 3|G-1|$ for $t \le -40$, together below 10^{-174}. The result is $2(\sum_k \nu_k - g - E) = 3.9796646\ldots$, a ball of radius below 10^{-24}. For the upper bound, ϑ and ϑ' are integrated in the same way at the ends of cells that cover $[-15, 0]$, and ϑ'' is enclosed over each cell. A cell is accepted when the Taylor bound of order two from one of its ends stays positive. Cells start at width $1/4$ and are bisected until they are accepted; 100 cells are needed. The smallest certified lower bound on a cell is 1.6×10^{-14}, and $\vartheta(0) = 9.94 \times 10^{-6}$. Below -15 the argument is in closed form. Put $w = -(t+g)$, $T = -g - 1/\nu_5$, $T' = -g - 2/\nu_5$ and $G_{\max} = 1 + \sum_k \nu_k/\sqrt{2\pi}$, so that $\tilde G \le G_{\max}$. The Mills ratio gives $\tilde G(t) - 1 \ge \varphi(w)(\nu_5 - 1/w)$ for $w > 0$, because $s_5 = -g$. Hence $\log \tilde G \ge 0$ for $t \le T$, and $\log \tilde G \ge (\tilde G - 1)/G_{\max} \ge \nu_5\varphi(w)/(2G_{\max})$ for $t \le T'$. From $\nu_1\varphi(t - s_1) \le \tilde G \le G_{\max}$ we also have $|\log \tilde G(t)| \le a_2 + t^2$ on $\mathbb{R}$, with $a_2 = \max(\log G_{\max},\, s_1^2 - \log(\nu_1/\sqrt{2\pi}),\, 0)$, where s_1 and ν_1 are the position and the mass of the first atom of $\tilde G$. Split $\vartheta(u)$ at T' and T, drop the nonnegative middle part, and use $(T+v)^2 \le T^2 + v^2$ and $\varphi(D_u + v) \le \varphi(D_u)e^{-D_u v}$ for $v \ge 0$, where $D_u = T - u > 0$. This gives $\vartheta(u) \ge \vartheta_+(u) - \vartheta_-(u)$ with

$$\vartheta_+(u) = \frac{\nu_5\, e^{-(u+g)^2/4}}{4\sqrt{\pi}\, G_{\max}}\,\Phi\Big(\sqrt{2}\,\Big(T' - \frac{u-g}{2}\Big)\Big),$$

$$\vartheta_-(u) = \varphi(D_u)\Big(\frac{a_2 + T^2}{D_u} + \frac{2}{D_u^3}\Big).$$

For $u \le 2T + g$ each factor of ϑ_-/ϑ_+ decreases as u decreases: the exponent $(u+g)^2/4 - (T-u)^2/2$ has the derivative $T + g/2 - u/2 \ge 0$ there, D_u grows, and Φ grows. Since $-15 \le 2T + g$ and $\vartheta_+(-15) > \vartheta_-(-15)$, with 3.2×10^{-15} against 1.2×10^{-15}, we get $\vartheta > 0$ on $(-\infty, -15]$. The dual value $2(\sum_k \nu_k - g) = 3.990007566$ is a ball of radius below 10^{-13}. The code of both computations and its output are in the supplementary material.

In the bulk $|x| \le A/2$ the output density equals e^{-C-h_1} up to a factor $1 + O(e^{-cA^{2/3}})$, which we prove (Corollary 1); we do not derive the strip law from it. Numerically the expansion agrees with this β to 5×10^{-12} on all 25 certified optimal states with A between 29.85 and 46.43, and six states computed afterwards gave the same value. For the NPMLE, $G \to 2D_N$ as $A \to \infty$, a defect D_N per end. For the LFP, fits on $m \le 40$ give the Bayes-risk deficit as $1 - B = \pi^2/(m+\delta)^2$ through order m^{-3}; the leading term π^2/m^2 is Bickel's [27], and δ widens the interval at each end. Two response constants enter the value laws of the Supplementary Notes: k_r, the response of the capacity strip to a linear reward on the input, and $D'(0)$, the derivative of the NPMLE defect when the data density has a small log-slope.

TABLE I
ONE-DIMENSIONAL EDGE CONSTANTS

constant	value	status
capacity strip β	3.98160794709	measured[a]
capacity reward response k_r	-0.63226	fitted[b]
NPMLE edge defect D_N	-0.304944111846	measured
NPMLE slope response $D'(0)$	$-0.041319 \pm 2 \times 10^{-6}$	post hoc[c]
LFP edge shift δ	3.34776	extrapolated[d]

[a] On 25 certified optimal states, to 5×10^{-12}. Proved bounds on its lower and upper limits as $A \to \infty$ are given in the text. [b] At $A = 16$ and 20.3; the $A = 16$ rows existed before registration; $A = 20.3$ replaced the registered $A = 20$, just past the count change at $A = 19.97$, where the solver stopped at residuals 10^{-5} to 6×10^{-3}, above the acceptance bar 10^{-10}; a fit with k_r alone gives -0.62970. [c] The registered extraction gave -0.041438; the corrected value was confirmed blind at one width inside the fitted range. [d] From fits on $m \le 40$; the registered check of the extrapolation at $m = 41$, 43, 45 is unscored.

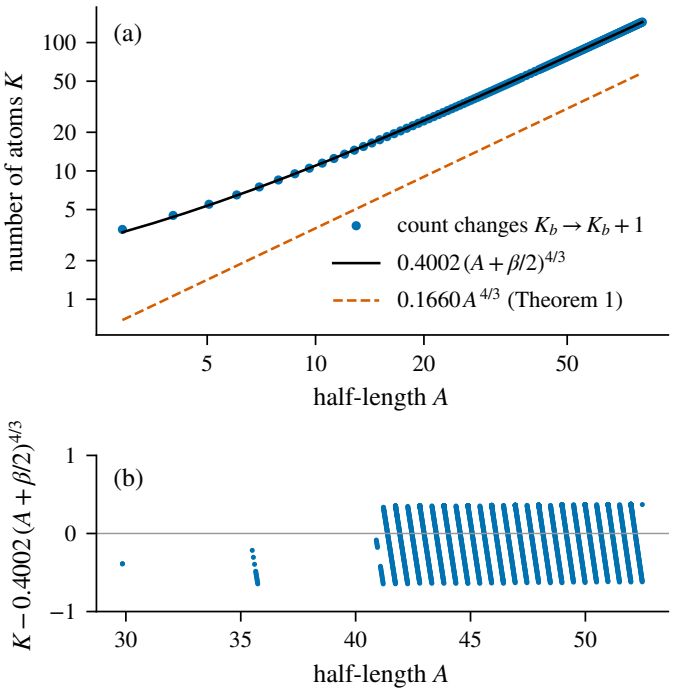


Fig. 1. The count law for the capacity on $[-A, A]$. (a) Every stored amplitude at which the number of atoms changes from K_b to $K_b + 1$, for K_b up to 144, drawn at $K = K_b + 1/2$. The solid line is the law $0.4002\,(A + \beta/2)^{4/3}$ of this section, and the dashed line is the asymptotic lower bound $0.1660\,A^{4/3}$ of Theorem 1. (b) The difference $K - 0.4002\,(A + \beta/2)^{4/3}$ at the 2666 stored optimal states with $A \ge 29.85$.

Numerically, the number of atoms grows like the $4/3$ power of the length, with the same constant in the three problems (Fig. 1). With L the half-length in noise units, the count is $K \approx 0.4002\,L^{4/3}$ plus terms of lower order. For the capacity we take $L = A + \beta/2$, the half-length with the strip of Table I; at all 2666 stored optimal states with $A \ge 29.85$ the difference $K - 0.4002\,L^{4/3}$ lies between -0.646 and 0.374. The constant $0.40024 = \frac{3}{2}\big(3/(16\pi^2)\big)^{1/3}$ follows from a local spacing law: at depth x from an end the atoms have density ρ with $\rho^3 =$

$3x/(16\pi^2)$, and integrating ρ over the depth $0 \le x \le L$ from each end gives the count. This law is the leading order of a bulk law that we obtained from a continuum phase condition for the first harmonic of the output density; the derivation is not rigorous, and we treat the law as a numerical one.

The exponent was found numerically by Mattingly et al. [13] and derived non-rigorously by Abbott and Machta [14], whose effective equation fixes a local density law. Read with one wall and with lengths in units of the noise standard deviation, their equation implies the count constant $\frac{3}{2}\,(27/(128\pi^6))^{1/6} = 0.3684$, which is 8% below ours. The ratio of the two constants is exactly $(6/\pi^2)^{1/6}$. A factor 2 between a full length and a half-length would change a count constant by $2^{4/3}$, so the difference is not one of the convention for the length. If their local law has the form $\rho^3 \propto x$ of ours, the difference is a factor $(6/\pi^2)^{1/2}$ in its prefactor. The constant 0.40024 and its agreement across the three problems come from our computations, which include forward tests, fixed before the computation, on twenty count changes for the LFP and for the NPMLE. Theorem 1 proves a lower bound of the same order $A^{4/3}$ for the capacity, with a smaller constant (Section III-B).

As L grows, the support changes at the center of the interval: an atom is born at the origin, and later the atom at the origin splits into a pair, so births and splits alternate. For capacity every local change of the support takes place at the origin, by Corollary 2 of [17]; that births and splits alternate is what our computations show, for capacity as for the LFP and the NPMLE. The first events are classical. The binary input is optimal up to $A = 1.665925641$ [9]; Sharma and Shamai, who characterized this transition, reported 1.671 [28]. The two-point prior is least favorable up to $m = 1.0567$ [4]. The single atom of the NPMLE splits at $R = \sqrt{3}$, where the variance of the uniform data equals the noise variance; this is the first critical temperature of deterministic annealing [1]. The Supplementary Notes use the whole sequence as the reference for births at the center of a ball.

B. The lower bound of Theorem 1

Theorem 1 concerns the exact optimizer, not an input within ϵ of capacity. The proof has four steps; Appendix A gives each in full.

1) The Kuhn–Tucker conditions bound the output density p above and below by constants times $c^* = e^{-C-h_1}$, with absolute constants, on the whole line.
2) A weighted identity, with the weight $\chi(x) = \exp(\Lambda_0(1 - x^2/A^2))$ and $\Lambda_0 = c_0 A^{2/3}$, and Plancherel's formula for Gaussian mixtures show that the weighted fitting error $E_2 = \int \chi\,(p - c^*)\log(p/c^*)\,dx$ is at most a constant times c^*, for every $c_0 < c_{0,\max} = (4\rho_1)^{-1/3}$.
3) A local lemma of Toeplitz type bounds the number of atoms in a window from below by the size of the fitting error relative to the window length.
4) The interval is tiled by windows of length $2A^{2/3}$ and the bounds are summed.

The only numerical input is the sign of one explicit function at $\lambda_1 = 1.1648$, evaluated in ball arithmetic (Lemma 3(c) in Appendix A). The constant 0.1660 is smaller than the numerical constant 0.4002 of Section III. The theorem is about the capacity and about a lower bound; it gives no upper bound and it does not cover the LFP or the NPMLE, for which the count is known numerically only. As a by-product, Corollary 1 shows that the output density is flat in the bulk, which is the property of the bulk that the strip law of Section III assumes; the law itself is not derived from it.

IV. Births on a line: the tower of events and the births law

A. The tower of events

Section III counts the atoms of the optimum at one amplitude. This section follows one optimum as the allowed amplitude grows. The amplitudes at which a new atom appears form the *tower of events*. We take the transmitter of Section II, with the channel $Y = X + Z$, unit Gaussian noise and the peak limit $|X| \le A$. Its best input is symmetric. It has atoms at $\pm u_j$ with masses w_j, two atoms at $\pm A$ against the wall, and, when the number of atoms K is odd, one atom at the origin. Smith proved that K is finite [2].

Along a branch of optima with fixed K, the positions and the masses move smoothly with A. A branch ends at an amplitude where the structure at the origin changes. For even K the branch ends in a *birth*: a new atom appears at the origin and K becomes $K+1$. For odd K it ends in a *split*: the atom at the origin separates into a pair and K becomes $K+1$. We write K_b for the number of atoms before the event. Barletta and Dytso proved that every change of the support happens at the origin, and that around every amplitude the count at all amplitudes close enough to it, below or above, is the count there or one more (Theorem 3 of [17]). Whether $K(A)$ can ever decrease is open since Sharma and Shamai [28]. This paper does not settle it. Section V-A reduces it to two statements.

We followed the optimum from the birth $K_b = 60$ to the birth $K_b = 82$. Each branch was computed in steps of 0.005 in A with a Newton corrector on the Kuhn–Tucker system at 200 bits or more, and the end of each branch was located by a secant method. This gives 23 events between $A = 41.2008363$ and $A = 52.4747205$. The amplitudes of the first events are 41.2008363, 41.7335208, 42.2640268, 42.7923985, 43.3186783 and 43.8429072 for $K_b = 60$ to 65; the last three are 51.4842403, 51.9802436 and 52.4747205 for $K_b = 80$ to 82. At every stored state the solver reports an optimal input, which includes the check that the optimality function stays below the capacity off the support. Optimality between two stored states, which lie 0.005 apart, is not tested. Table II lists the first events of the tower.

A second, independent code extends the tower. It uses ball arithmetic (192 bits), reproduced four events of the tower and was then registered for use before any event above $K_b = 82$ existed. With it we located the events $K_b = 83$ to 144. All 22 events $K_b = 61$ to 82 were also measured again with it. The shifts against the first instrument are below 5.0×10^{-13} in A, with one exception: $K_b = 73$, where the stored amplitude was off by 1.1×10^{-11} and the corrected value is used below.

TABLE II
AMPLITUDES OF THE FIRST EVENTS OF THE TOWER (CAPACITY, UNIT NOISE). K_b IS THE NUMBER OF ATOMS BEFORE THE EVENT.

K_b	A	kind	K_b	A	kind
60	41.2008363	birth	66	44.3651245	birth
61	41.7335208	split	67	44.8853686	split
62	42.2640268	birth	68	45.4036762	birth
63	42.7923985	split	69	45.9200831	split
64	43.3186783	birth	70	46.4346238	birth
65	43.8429072	split	71	46.9473315	split

The families keep a clear shape. At the end of an odd branch the second derivative of the optimality function at the origin passes through zero, and at the end of an even branch the optimality function at the origin passes through the capacity level. The split is a pitchfork: the separation of the pair grows like the square root of $A - A_s$, where A_s is the amplitude of the split, with a fitted coefficient between 0.2590 and 0.2455 along the tower. The birth is a crossing of the mass of the new atom through zero, upward in A.

B. The births law

The births law is the count law of Section III read at the events. There the atoms at depth x from an end have, to leading order, the density ρ with $\rho^3 = 3x/(16\pi^2)$, and the integral of ρ from each end gives the count. Here the density is taken from the optimum on a half line, and the atoms counted from both walls must reach the number of atoms before the event, up to a small correction.

At what amplitude does the next event happen? Near the wall, the optimum of the finite problem looks like the solution of a problem on a half line with a single wall. For the capacity it is the following. The atoms lie at depths $d \geq 0$ below a wall at $d = 0$, which carries one atom. The output density $P(y) = \sum_j w_j \varphi(y - d_j)$ is normalized so that a unit mass per unit length gives $P = 1$ in the bulk, and $g(x) = -\int \varphi(y - x) \log P(y)\, dy$ plays the role of the optimality function, with the level 0. The chain is truncated at a depth D. The atoms with $0 < d \leq D$ are free, with $g = 0$ and $g' = 0$ there. In a zone of width B below them the positions follow the bulk law of Section III and the masses solve $g = 0$. Beyond it the atoms are fixed on the bulk law with mass $1/\rho$, and the wall atom solves $g = 0$. The known events below use the solution with $D = 110$ and $B = 14$, the later blocks the one with $D = 134$ and $B = 20$. Let $N_{\mathrm{half}}(a)$ count the atoms of this half-line solution within distance a of the wall. Let $\rho = dN_{\mathrm{half}}/da$ be the density of atoms and $v = 4\pi\rho$. The *births law* says that an event at the origin happens at the amplitude A where

$$2\, N_{\mathrm{half}}(A) \;=\; K_b + \tfrac{1}{6} + \delta_b(\rho), \tag{4}$$

with ρ evaluated at A. In words: the new atom appears when the atoms counted from the wall, doubled to cover both sides, reach K_b plus a small correction. The form of the law has no free parameter and it uses no data of another problem. One law covers births and splits. The accuracies below depend on choices that were made on known events: the order of the series, the degree of the spline for N_{half} and the depth to which the half-line problem is solved.

The correction δ_b is a series in $1/v^2$ without a term of order $1/v^2$,

$$\delta_b \;=\; e_2/v^4 + e_3/v^6 + e_4/v^8 + \cdots ,$$

and its coefficients come from the problem at the origin with the density held fixed. They are the same for the capacity, the NPMLE and the immune problem, which share the Gaussian kernel and a uniform weight. The immune problem is the member $k = 2$ of a family of problems whose optimality function is the integral of the Gaussian kernel, a weight and the output density to the power $-k$ (Supplementary Notes); the NPMLE is $k = 1$. The capacity is the limit $k \to 0$ of a companion family in which $(p^{-k} - 1)/k$, with p the output density, replaces the weighted power and the atoms are confined to $[-A, A]$ with one atom at the bound (Supplementary Notes). The first coefficients have closed forms,

$$e_2 = \tfrac{4}{3} + \tfrac{8\pi^2}{45} = 3.08793,$$
$$e_3 = \tfrac{176}{15} + \tfrac{32\pi^2}{9} + \tfrac{64\pi^4}{1575} = 50.7835,$$
$$e_4 = \tfrac{23792}{315} + \tfrac{8416\pi^2}{135} - \tfrac{4544\pi^4}{2025} + \tfrac{2176\pi^6}{33075} = 535.477,$$

and $e_5 = -4254.456$, also of this form. Closed forms of the same kind are known through e_{10}, and numerical values through e_{20} are stored in the repository. The coefficients e_2 to e_4 are positive and e_5 to e_{20} are negative. The derivation reads the Kuhn–Tucker condition at the origin as a heat flow of the phase of the atoms. Each order in $1/v^2$ gives a linear problem that is solved in polynomials and in $\coth$. It rests on stated assumptions, and its aliasing terms, of order $e^{-\pi^2\rho^2}$, are not bounded. The coefficients grow in absolute value like $n!$ times about 4.4^n, so the series is asymptotic. A model of the ratios e_{n+1}/e_n fitted to $n = 10$ to 17 predicted e_{19} and e_{20} to 0.04% and 0.11% before they were computed.

a) Tests: We separate readings of events that were known when the law was fixed from tests registered before the data. All deviations are $A - a_{\mathrm{pred}}$ in units of 10^{-9}, with the series cut at order 16 and a quintic spline for N_{half}.

- *Known events.* The law reproduces the 20 capacity events $K_b = 61$ to 80 to within 0.09, and the ten splits among them to within 0.016. It reproduces 13 NPMLE transitions $K_b = 54$ to 66 to within 0.07. The largest deviations are births, and each order of the series up to order 10 lowers the deviation by a factor 2.5 to 3.
- *Blind, capacity.* Two tables of predicted amplitudes were fixed before the split $K_b = 81$ and the birth $K_b = 82$ were located. The first table comes from a variant of the law that multiplies the derived terms by a ratio curve measured on the immune problem (bar 0.5). The second is the derived series at order 12 (bar 0.3). The first value of the amplitude of the split carried a termination error of 0.109 from the routine that located the event. It deviates by $+0.119$ and $+0.098$ from the two tables, and by $+0.010$ and -0.011 after the correction. The birth deviates by $+0.005$ and -0.006. All of these are inside their bars. The same two tables were then tested on later events up to $K_b = 94$. The first table failed its bar at the

birth $K_b = 94$, with a deviation of $+0.525$ against 0.5. The derived series, tested on the same events, deviated by $+0.014$ at $K_b = 93$ and $+0.020$ at $K_b = 94$.

- *Blind, NPMLE.* Of the 34 transitions $K_b = 67$ to 100 only 13 were located under the registered definition. They lie within -0.043 and $+0.037$ of the derived-series table (bar 0.3). Against the table of the first variant one of them, $K_b = 93$, deviates by $+0.523$ and misses its bar 0.5.
- *Blind, later capacity events.* Three blocks of eight events were predicted from converged solutions of the half-line problem before any of them existed: $K_b = 115$ to 122, 123 to 130 and 131 to 138. All three blocks passed their registered bars, with largest deviations 0.0338, 0.1517 and 0.6399. The bar of an event is the larger of 0.05 and twice the spread of the half-line solutions, which grows with K_b.

A finer version of the law replaces the half-line density by a series for the density of the atoms. With the exact center series and the bulk law carried to the order 18, it predicts the events $K_b = 61$ to 82 to within 2.5×10^{-13}, the events $K_b = 83$ to 130 with a root-mean-square deviation of 4.8×10^{-14}, and the events $K_b = 131$ to 143 to within 1.1×10^{-13}. This is not a blind test. The orders of the series were chosen on the lower events, and among the later events the orders 17 to 22 cannot be told apart.

b) What failed: Two registered tests of a first version of the law, which cut the series at e_5 and took the density from a fit to the half-line atoms, failed by the letter; Section IX lists them. The Supplementary Notes give their values and the two instrument choices to which the law is sensitive.

A remark. The smallest gap between two consecutive predicted amplitudes is 0.455. For $41.2 \le A \le 67.6$ the law therefore places births and splits one after the other. This says nothing about an event that the law does not predict and that would lower K. The missing hypothesis is that the list of events is complete, which is the content of statements (S1) and (S2) of Section V-A and is not proved.

V. Outward motion and monotonicity

This section is about the motion of the atoms as the amplitude grows. Section V-A reduces the question whether the number of atoms can decrease to two statements, (S1) and (S2), about the innermost atoms. Section V-B shows that the outward motion of the positions, which gives (S2), follows from the total positivity of one matrix. It proves that property for the Gaussian part of the matrix and, by a computer-assisted proof, for the whole matrix at the optimal states of Proposition 1.

A. Can the number of atoms ever decrease?

Sharma and Shamai conjectured in 2010 that the number $K(A)$ of atoms of the capacity-achieving input never decreases when the amplitude A grows [28]. Barletta and Dytso proved that every change of the support happens at the origin [17]. The origin atom can appear or disappear, and the origin can split into a symmetric pair or such a pair can merge there. Around every amplitude, the count at all amplitudes close enough to it, below or above, is the count there or one more (Theorem 3 of [17]). They state that global monotonicity is open, and they report numerical evidence, for A up to about 8, that the sorted masses are majorized along A. This subsection reduces the question to two statements about the innermost atoms and reports the evidence for them. It proves both statements at the optimal states of Proposition 1 and neither in general.

a) The reduction: Inside a family K is constant. By Theorem 3 of [17] the positions and masses of the nonzero atoms vary continuously along a family, and a family can end only at the origin. It can end in four ways. A fifth way, for odd K the innermost pair merging with the existing center atom ($K \to K - 2$), is excluded: at amplitudes close to the end at most two support points lie near the origin (the proof of Theorem 3 and Corollary 2 of [17]). A birth or a split raises K. The loss of the origin atom (odd K) or the merger of the innermost pair at the origin (even K) lowers K. So $K(A)$ is nondecreasing if the last two never happen. They would need the mass of the center atom to fall to zero, or the position of the innermost atom to fall to zero, as A grows. Hence $K(A)$ is nondecreasing if on every family

(S1) the mass of the center atom increases with A (odd K), and

(S2) the position of the innermost atom increases with A (even K).

This reduction is a proved implication, given that theorem and the continuation of each family by continuity up to its end. The statements (S1) and (S2) are not proved in general. At the optimal states of Proposition 1 both hold: (S2) by the outward motion, and (S1) by Step 6 of Appendix D.

b) The evidence: The two statements hold in every stored pair of states of the same family in four problems: the capacity (2639 pairs inside a family), the immune problem (1086), the NPMLE problem (676) and the least favorable prior (141). The smallest relative growth rate is 0.15 per unit of A, about eight times the rate $1/(A + a_w)$ at which the global scale falls at A about 50, where a_w is the shift of the scaling law of the wall masses stated next. The capacity pairs come from the stored states of the families $K = 25$, 26, 40 and 60 to 83, which reach $A = 52.5$. For small A the two statements were checked on new double-precision families ($A = 1$ to 15.8 and $K = 2$ to 18, with 2422 pairs, not certified) and on 27 certified states in the gap $A = 16.3$ to 19.9 (19 pairs). No pair violates them, and the relative growth rates are at least 0.20. The masses of the atoms near the wall obey a scaling law that holds to high accuracy and is not exact: $w_j(A)\,(A + a_w)$ is constant inside a family, with $a_w = 1.990803974$ for the capacity (2285 pairs, the edge atom to 4×10^{-10}). This shift equals $\beta/2$ of Table I to the digits shown. It is a measurement, and we have no derivation of it.

The majorization conjectured by Barletta and Dytso holds for the capacity on all 2665 consecutive stored pairs, 26 of them across a change of K and 2639 inside a family, with a largest violation of 1.6×10^{-15}, and for the immune and NPMLE problems. It fails for the least favorable prior, where 149 of 181 pairs violate it. Majorization is therefore not a general route, and the reduction above does not need it.

c) The flow at the end of a branch: The flow of the atom at u_j is du_j/dA. The smallest flow of a branch is attained at its last state, just before the event. Take the density ρ of the births law at the event. The product of the smallest flow and ρ^2 is nearly constant in each parity. It is 0.1745 for odd K and 0.0879 for even K, with a scatter of 0.4% for $K = 63$ to 72 and 0.5% for $K = 73$ to 79. The families $K = 73$ to 79 were predicted before they were computed. The flows of the ten innermost atoms at the event fall on one curve of the scaled position $\xi_j = u_j h/(4\pi)$, with $h = u_2 - u_1$, in the capacity, the immune and the NPMLE problems. The curve is $f(\xi) = \coth(\pi\xi) - \pi\xi/\sinh^2(\pi\xi)$ with no fitted number, and it agrees with the data to an rms of 0.0017 to 0.0026. It follows from the linearised phase condition of the sharp center solution under five stated assumptions. The assumptions are a dense lattice, an unchanged phase along the branch, minimal singularities, a sign-function limit far from the center, and boundedness of the profile in a strip. The curve implies that the logarithmic flow $(du_1/dA)/(u_1 h)$ of the innermost atom tends to $1/6$. The data do not decide this: extrapolations give values between 0.1643 and 0.1675. The full set of tests, including the first correction in $1/\rho^2$, is in the repository.

d) Where the difficulty sits: Near the end of a family the growth of the innermost quantity competes with the decay of the global scale, so the sign is not dominated by one term. The implicit-function system shows where the competition sits. Write the Kuhn–Tucker system of Section V-B as $F(z; A) = 0$, with the rows $D'(u_j)$, $D(u_1) - D(A)$, $D(u_j) - D(u_1)$ for $j \geq 2$, $D(0) - D(u_1)$ for odd K, and the normalization of the total mass. Then $dz/dA = -J_F^{-1} b_F$, with J_F the Jacobian of F in z and $b_F = \partial F/\partial A$. The flow of the innermost unknown, u_1 for even K and the center mass for odd K, is the sum of one term per row, $c_k = -(J_F^{-T} e)_k \, (b_F)_k$, where e selects that unknown. With Σ_+ the sum of the positive terms and Σ_- minus the sum of the negative ones, the flow is $\Sigma_+ - \Sigma_-$, with Σ_+/Σ_- between 1.050 and 1.093 in 47 certified states ($K = 19$ to 81). Three atoms next to the wall carry 98 to 99 percent of the terms, and their depths below the wall are constants. So a proof needs $\Sigma_+ > \Sigma_-$ for the response of the edge region to the motion of the wall. The margin is 5 to 9 percent and falls slowly with K.

We measured the margin on the rows $K = 83$ to 130. The smallest value of Σ_+/Σ_- over the 48 rows is 1.046845, and the margin is positive on every row. Its decrease with K is described by a series in $K^{-1/2}$ that was fitted on the rows $K = 83$ to 114. The series predicted the rows $K = 115$ to 122 and 123 to 130 before they were made, and both blind tests passed. With four coefficients the limits of Σ_+/Σ_- are 1.0344 for even and 1.0346 for odd rows. A fifth coefficient brings both parities to 1.03440. For a second ratio, computed from the edge atoms alone, the limit is 1.05416. The limits are extrapolations of data to $K = 130$ and lie above one. A fit with a free exponent on the rows to $K = 108$ puts them higher by 0.3 to 0.5 percent. The data cannot tell whether the margins stay positive for very large K. Two attempts to compute the edge region exactly failed as formulations: a truncated edge block of the n atoms nearest the wall converges only like $n^{-0.85}$, and a geometric closure with exact boundary ratios does not remove that error.

The total positivity of Section V-B is the candidate tool. It gives the outward motion of the positions at the 91 optimal states of Proposition 1, and the one-minimum lemma of Appendix B (one local minimum of $D - C$ between consecutive atoms) limits the oscillations of the optimality function. Neither gives $\Sigma_+ > \Sigma_-$. This does not move the upper bound $K = O(A^{4/3})$, which is also open. The best proved lower bound is Theorem 1.

B. Why the atoms move outward: total positivity

Do the atoms of the optimum move away from the origin as the amplitude grows? If they do, the innermost pair of an even branch cannot merge at the origin, which is statement (S2) of Section V-A. This subsection shows that the outward motion of the positions follows from the total positivity of one matrix. The mass of the center atom, statement (S1) of that subsection, is not covered by it; at the exact states of Proposition 1 it is proved by a separate enclosure (Appendix D). We prove the total positivity of the part of the matrix built from the Gaussian kernel, and, by a computer-assisted proof, of the whole matrix at the 91 optimal states of this subsection. We do not prove it in general.

a) The matrix: Take a branch of optima with K atoms, m positive atoms at $0 < u_1 < \cdots < u_m < A$ with masses w_j, and the wall atom at A with mass w_e. The atoms at $-u_j$ and $-A$ have the same masses, and a center atom for odd K has the mass w_0. Let $D(x)$ be the relative entropy of the unit normal centred at x from the output density Q. The Kuhn–Tucker conditions are $D'(u_j) = 0$ and $D(u_j) = D(A)$ for $j = 1, \ldots, m$, and $D(0) = D(A)$ if there is a center atom. Differentiating these along the branch and eliminating the masses leaves a matrix S of size m, the reduced Hessian of the capacity in the positions, and a forcing vector b, so that $du/dA = -S^{-1} b$. With $E = \operatorname{diag}((-1)^{p-1})_{p=1}^{m}$ and $\sigma = (-1)^{m+1}$ we put $T = -ESE$ and $v = \sigma E b$. Then $du/dA = \sigma E \, T^{-1} v$.

Lemma 1. *If T is strictly totally positive and $X = [\, T \mid v \,]$ has all its maximal minors positive, then $du_i/dA > 0$ for every i.*

Proof. By Cramer's rule the component i of $T^{-1} v$ is $\det(T$ with column i replaced by $v)/\det T$. Moving v to the last column costs the sign $(-1)^{m-i}$, so this component is $(-1)^{m-i}$ times a maximal minor of X divided by $\det T$. Both are positive, and the sign of du_i/dA is $\sigma(-1)^{i-1}(-1)^{m-i} = \sigma(-1)^{m-1} = +1$. □

The lemma turns a question about the sign of an alternating sum into a question about a matrix. The rest of this subsection is about that matrix.

b) What is proved: Let $\langle f, g \rangle = \int fg/Q$ over the real line. The *value functions* are the output bumps $\varphi_j = \varphi(\cdot - u_j) + \varphi(\cdot + u_j)$ of the atoms, and the *derivative functions* ψ_p are their derivatives in the positions, weighted by the masses. The value functions belong to the *value nodes* $u_1, \ldots, u_m$ and A, with $\varphi_A = \varphi(\cdot - A) + \varphi(\cdot + A)$, and for odd K also to the node 0, with $\varphi_0 = \varphi$; each derivative function has the node of its atom. The *mass directions* $\chi_k = \varphi_k - \varphi_A$, and for odd

K also $\chi_0 = \varphi_0 - \varphi_A/2$, are the changes of the masses that keep the total mass equal to one. Let N be the Gram matrix of the derivative functions after removing their projection on the span of the mass directions. This is a Schur complement of the full Gram matrix.

Theorem 2. *For every configuration $0 < u_1 < \cdots < u_m < A$ with positive masses, whether or not it solves the Kuhn–Tucker system, the matrix $E'NE'$ of order $m+1$ is strictly totally positive. Its rows and columns are indexed by the positive atoms and the wall atom, $E' = \mathrm{diag}((-1)^{c(p)})$, and $c(p)$ counts the value nodes at or below the node of p.*

The proof is in Appendix C. Its tools are classical: the reflected Gaussian kernel is extended totally positive, the basic composition formula of Karlin [29] turns a Gram minor into an integral of a product of two positive determinants, and Schur's determinant formula handles the projection. A numerical check agrees: at 40 random configurations with $m = 5$ to 9 and no optimality condition, all 2930 initial minors are positive. For a square matrix this is enough for strict total positivity [30].

At a stationary configuration the reduced Hessian splits as $S = S_G + \mathrm{diag}(D''(u_p))$, where $S_G = \mathrm{diag}(c)\,N$ with $c_p = -1/(2w_p)$ is the Gram part. Step 5 of Appendix C derives this identity, and it was confirmed to between 10^{-21} and 10^{-36} at six states. The diagonal term is itself a Gram sum, $D''(u_p) = \frac{1}{2}\sum_q \tilde{w}_q \langle \psi_p^o, \psi_q^o \rangle$, of the odd derivative functions $\psi_q^o = \partial_u[\varphi(\cdot - u) - \varphi(\cdot + u)]$ at $u = u_q$, at $u = A$ for the wall atom and $\psi_0^o = 2y\varphi(y)$ for a center atom, with the weights $\tilde{w}_q = w_q$, w_e and $w_0/2$. Step 6 of Appendix C proves this second identity at every configuration, with or without stationarity, and it holds to 8×10^{-60} at 40 random configurations. So both parts of the Hessian are built from one family of Gaussian functions, and, as an observation, the sign of $-D''$ that the optimum needs comes from the non-uniformity of the lattice of atoms. Let $b_G = c\,N_{\cdot A}$ be the Gram part of the forcing, and let $T_G = -ES_GE$ and $v_G = \sigma E b_G$. Then $X_G = [\,T_G \mid v_G\,]$ equals $\mathrm{diag}(-c)$ times the first m rows of $E'NE'$ in the columns $1, \ldots, m, A$, because $c_p < 0$ and the signs satisfy $E_p = (-1)^{K+1}E'_p$ for $p \le m$ and $E'_A = (-1)^K\sigma$; here N in S_G and b_G means its block of the positive atoms and its column A. The Gram part of T and of X is therefore strictly totally positive at every stationary configuration. The forcing has a second part outside the Gram part: the rows $D(u_j) = D(A)$ are evaluated at the wall, which moves with A, so their derivative carries the term $D'(A)$, and $D'(A) > 0$ (Appendix B). What remains to be shown is that the diagonal term and this part of the forcing do not destroy the total positivity.

c) What is checked: A matrix of any shape, square or rectangular, is strictly totally positive when its initial minors are positive [30, Thm. 4.1]. For X with m rows these are $m(m+1)$ minors. We computed all of them at 1200 bits, with the radii of the balls kept, at states polished to a residual below 10^{-120}. At the older accuracy of the stored states, 10^{-39}, the corner minors of size 10^{-241} to 10^{-364} change sign with the noise, so the precision matters. At every one of 94 distinct states of the capacity all initial minors of X are certainly positive. Of these, 91 are optimal states of the families $K = 26$, 40, 50, 60 to 66 and 70, with 12 to 34 positive atoms. Three are stationary states beyond the end of their family, at $K = 40$, 70 and 80, the last with 39 positive atoms. By Lemma 1 the positions move outward at the stored state of each of them. The balls above certify the minors of the matrix built from each stored state, which only approximates the exact optimal state. The next proposition removes that gap.

Proposition 1 (computer-assisted)**.** *Let A be the amplitude of one of the 91 optimal states above, and let K be its number of atoms. The capacity-achieving input at amplitude A has K atoms, and its positions and masses lie within 10^{-110} of those of the state polished from the stored one; both states are in the supplementary material. All initial minors of its matrix X are positive, and its positive atoms move outward as A grows: $du_i/dA > 0$ for every i. If K is odd, the mass of its center atom grows as A grows: $dw_0/dA > 0$. At the three stationary states beyond the end of their family, the Kuhn–Tucker system has exactly one solution in a box around the polished state, and all initial minors of X are positive in that box. The box has radius 10^{-110} at $K = 40$ and $K = 70$, and radius 10^{-150} with the grid step 0.005 at $K = 80$.*

The proof is computer-assisted and is given in Appendix D. It finds a small box that contains the exact solution of the Kuhn–Tucker system, with the integrals of the channel and not their trapezoid sums, and then shows in ball arithmetic that this solution satisfies the optimality condition on the whole interval.

The property is fragile. Moving the positions by one percent of a spacing, or the masses by one percent, turns 32 to 54 percent of the initial minors negative. The region of total positivity has a relative size of about 10^{-25} at the one polished state where we measured it, in one random direction. Here the relative size is the displacement, with positions in spacings and masses in their own size, at which a minor first turns negative. The box of radius 10^{-110} in Proposition 1 lies far inside this region. The same extension is strictly totally positive on 14 rows of the NPMLE functional and on 15 rows of the immune functional.

d) The margin: Multiply the diagonal term by a factor t, and let t^* be the smallest t at which the decisive minor of X, the one that fixes the sign of du_1/dA, turns negative. Then $t^* > 1$ at every state examined. With J positive atoms, $J(t^* - 1)$ is nearly independent of K at the first, median and last state of a family, and it lies between 4.0 and 4.8. The values for $K = 68$ to 71 were predicted inside registered bands before the states were read. So $t^* - 1 \approx a/J$ with a between 4.0 and 4.8: the margin closes like $1/J$ and stays positive at all states examined, with J up to 34 in optimal states and 39 at one stationary state. The decisive minor is a polynomial of degree $m-1$ in t. Its roots are all real, simple and positive at 35 states of eleven families, and they grow almost geometrically (Fig. 2). A proof of $t^* > 1$ for all K needs a bound on such a root, or on a power sum of their reciprocals. We do not have one, and every certificate we found has a margin proportional to the innermost flow du_1/dA itself. Whether the scaling persists for J far above 40 is not known. If it does, the margin $t^* - 1$ tends to zero as J grows, and

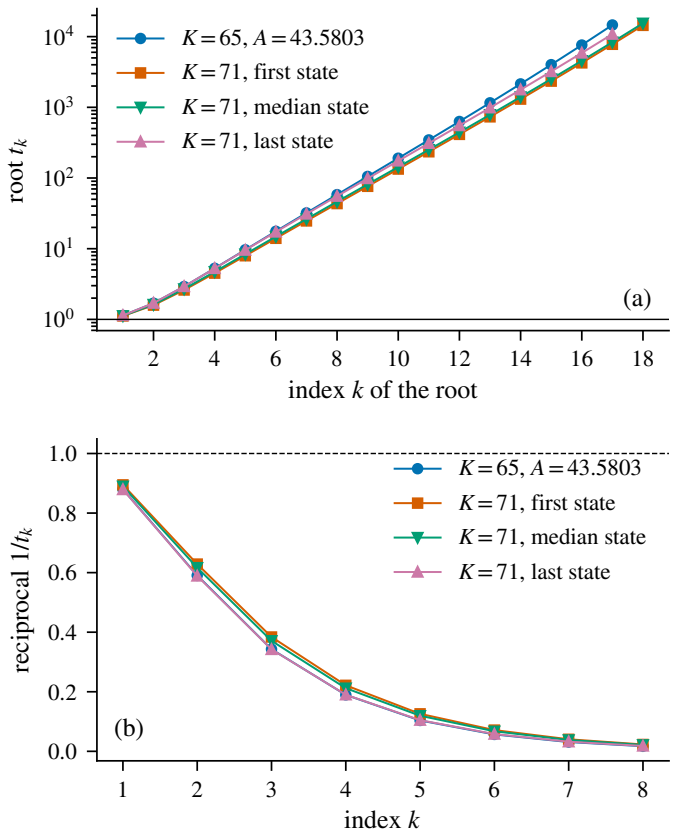


Fig. 2. The roots of the decisive polynomial at four stored states of the capacity: one state of $K = 65$ and the first, median and last state of $K = 71$. (a) The roots t_k below 20073.7 on a logarithmic scale; they grow almost geometrically. The polynomial has degree $m - 1$, 30 for $K = 65$ and 33 for $K = 71$, and all its roots are real and positive. (b) The reciprocals $1/t_k$ of the first eight roots. All are below one.

the total positivity of the actual matrix becomes marginal for large K; the route of this section then needs the bound on the roots mentioned above.

VI. Closed Curves: The Ring Copies the Formal Density

A. Two curves and two problems

Let $d = 2$ and let the input be confined to a closed curve near a circle, or to the region the curve bounds. We use two families,

$$z(t) = Re^{it} + \eta\,(e^{-it} + \lambda e^{3it}), \quad R = r_p - \tfrac{\epsilon}{2}, \quad \eta = \frac{\epsilon}{2(1+\lambda)}, \tag{5}$$

so that $z(0) = r_p$ and $\epsilon = r_p - r_m$. With $\lambda = 0$ the curve is the ellipse $(r_p \cos t, r_m \sin t)$; with $\lambda = 0.02$ it is a curve we call the mixed curve. Problem P1 confines the input to the filled region and P2 to the curve, and an optimum of P2 is an optimum of P1 exactly when its optimality function also stays below C inside. For the ellipse with $r_p \le \sqrt{2}$ the two problems have the same optimum (Theorem 1 of [18]). Both curves are symmetric under $x_1 \mapsto -x_1$ and $x_2 \mapsto -x_2$, and so are their optima. A ring therefore consists of the pair X at the ends of the major axis ($t = 0, \pi$), the pair Y at the ends of the minor axis ($t = \pm\pi/2$), and orbits of four atoms at $t = \pm a, \pi \pm a$.

B. Para-orthogonal quadrature

The description borrows from the theory of orthogonal polynomials on the unit circle [31], [32]. Let $\alpha_0, \alpha_1, \ldots$ be the Verblunsky coefficients of the measure $f(t)\,dt/2\pi$, where f is the formal density; they are defined by the Szegő recursion of the monic orthogonal polynomials Φ_n of that measure, $\Phi_{n+1}(z) = z\Phi_n(z) - \alpha_n \Phi_n^*(z)$, with $\Phi_n^*(z) = z^n \overline{\Phi_n(1/\bar{z})}$ the reversed polynomial, as long as $|\alpha_j| < 1$. They are real by the symmetry, and $\alpha_j = 0$ for even j because f has only even harmonics. For $\tau \in \{+1, -1\}$, the K-node para-orthogonal rule replaces α_{K-1} by τ. If $|\alpha_j| < 1$ for $j \le K-2$, the rule exists with K distinct nodes e^{it_k} on the unit circle and positive weights, and it reproduces the moments of orders $1, \ldots, K-1$. The Verblunsky coefficients of a signed f need not lie in the unit disc; we use the rule only while they do. We place its nodes on the curve at $z(t_k)$. Because f has only even harmonics, the layout of the nodes is fixed by τ and K. For $\tau = +1$ there are nodes at $t = 0, \pi$ (the X pair), and at $t = \pm\pi/2$ (the Y pair) exactly when $K \equiv 0 \pmod 4$. For $\tau = -1$ there is no X pair, and there is a Y pair exactly when $K \equiv 2 \pmod 4$. For a ring with masses m_k at angles t_k, where $\sum_k m_k = 1$, write $b_j = 2\sum_k m_k \cos j t_k$ for its harmonics. The K-th moment of a measure depends on α_{K-1} only through the term $\alpha_{K-1}\prod_{j<K-1}(1-\alpha_j^2)$, and the rule differs from the measure only in that coefficient. With the factor 2 that turns moments into cosine coefficients, this gives, for the rule's own harmonic of order K,

$$c_K - b_K = 2(\alpha_{K-1} - \tau) \prod_{j<K-1} (1 - \alpha_j^2), \tag{6}$$

which we also checked to twelve digits on computed rules.

Conjecture 1 (copying)**.** *Take the formal density of order $M = K/2$. While $|\alpha_j| < 1$ for $j \le K-2$, the optimal K-atom ring is the K-node para-orthogonal rule of that density, up to copying errors $h_j = b_j - c_j$ ($j \le K-2$). Along the optimal rings, whose number of atoms grows as $\epsilon \to 0$, the error h_j tends to zero for every fixed j. It has $\tau = +1$ if it carries the X pair and $\tau = -1$ otherwise.*

Conjecture 2 (loss)**.** *The K-atom ring stops being optimal near the ϵ at which $\alpha_{K-1} = \tau$, that is, where $b_K = c_K$. The kind of the event (an atom born or an atom split) and its place (the X or the Y pair) follow from the layout rule and the sign rule of Section VI-C.*

We state both at the precision of our computations, so that another group can falsify them. On the curves of Table III with $r_p \le 1.2$, the loss event lay within 0.28% of the point where $b_K = c_K$, and in the NPMLE, LFP and annealing events of Table IV within 0.61%; on the ellipse $r_p = 3$ the miss was about 1% and fell with K. The linear-response form of Section VI-C is a sharper statement: where it was computed, the loss lay within 10^{-4} (relative) of its prediction, and it predicted the copying error h_{K-2} to within 2.2%.

Solved rings copy the density closely. On the ellipse with $r_p = 1.3$ and $K = 12$, the errors $|h_j|$ are at most 4×10^{-6} for $j \le 8$, near the accuracy of the solver, and 6.2×10^{-4} at $j = 10$. The solved rings with $K = 10$ and 12 there have $h_{K-2} = -9.19 \times 10^{-4}$ and -6.21×10^{-4}, against -9.32×10^{-4} and -6.35×10^{-4} from the linear response of Section VI-C.

C. Why the ring copies, and the offset of the loss

The optimality function depends on the input only through the output density, and harmonic j of the input reaches the output density at radius r through the factor $I_j(rR)/I_0(rR) \approx (rR/2)^j/j!$. For a ring P near the density, therefore, $F(t;P) = F(t;f) + \sum_{ij} L_{ij} h_j \cos it$ up to second-order terms, with L nearly diagonal and $|L_{jj}|$ falling factorially in j. A stationary ring makes $g = F - C$ vanish with zero slope at each of its K atoms. Truncate g at harmonic K: a trigonometric polynomial of degree K that vanishes to second order at the K atoms is a multiple of $\Pi(t) = \prod_k (1 - \cos(t - t_k))$, so $g = \gamma\,\Pi$ for one number γ. Matching the harmonics below K then gives $h_j = 0$ for $j \le K - 2$ at leading order: $K/2 - 1$ equations for the $K/2 - 1$ free masses and angles, which is the para-orthogonal rule.

The loss condition $b_K = c_K$ tests a mean, while a ring is lost when a maximum of g reaches zero. Keeping the harmonics of g up to order $K + 4$ and expanding them linearly about the formal density gives $g = \Pi q$, with q a trigonometric polynomial of degree 4; the linear-response event is the ϵ at which $\max_t q = 0$. Since $\max q \ge q_0$, the mean of q over t, and the copying condition is $q_0 = 0$ to leading order, the event comes before the copying event as ϵ decreases: the offset is positive, as all six linear-response cases of Table III show.

The sign of the second harmonic of q gives the place. It equals $-\operatorname{sgn}(c_K c_{K+2})$, so when c_K and c_{K+2} have the same sign the maximum of q is at the Y pair, and otherwise at the X pair; an atom there splits, and a gap there gives a birth.

This argument is perturbative. We introduced the correction after the fact, on recorded events, before any blind test that uses it was registered.

D. The record

Each predicted value in Table III was appended to the log before any solve of the ring in question. The solves then tracked the optimum in ball arithmetic and bisected the first loss of the optimality condition. On the ellipse $r_p = 1$ the rule alone placed every event within 0.015%, and on the mixed curve within 0.28%. On the elongated ellipse $r_p = 3$ it missed by about one per cent, continuing a decreasing sequence of misses at smaller rings there. With the linear-response correction, computed for six events, the misses lie between 10^{-7} and 9.5×10^{-5}. Every registered kind and place was observed, except the kind of the 36-ring's loss. In the first invalid state the optimality function is positive in the gaps next to the empty Y position and largest at the vertex. That fits the predicted birth, but it does not separate the birth from a split of the neighboring atoms. The rings of 32 and 34 atoms on the ellipse $r_p = 1$ carry no X pair, and Section VII certifies one of them. The two valid 36-ring states we solved, at $\epsilon = 0.0310$ and 0.0308 in 448-bit arithmetic, carry none either: their atoms nearest the end of the major axis sit at $t = \pm 4.97°$ and $\pm 4.96°$, and $D_\pi - C$ at the end is -2.6×10^{-56} and -3.8×10^{-57}.

The same rule was tested in advance in the other two problems, on the ellipse $r_p = 1$ (Table IV). Five ring losses of the NPMLE with a Gaussian source ($s = 2$) and of the LFP lay within 0.61% of the rule alone, and the four for which the linear-response correction was registered lay within 3.5×10^{-6} (relative) of it. Deterministic annealing gives a second path through the NPMLE. At fixed shape ($r_p = 1$, $r_m = 0.7$, $s = 2$) the temperature T is lowered, and the problem at temperature T is the NPMLE with noise of standard deviation $\sqrt{T/2}$ [1]. Its first five ring losses on cooling lay within 0.37% of the rule. Every registered kind and place in these ten events was observed. With Table III, twenty-four ring losses predicted in advance in the three problems fell inside their bands.

E. The count as the curve becomes a circle

Read as a count, Conjecture 2 says that the optimum has $K = j^* + 1$ atoms, where j^* is the smallest odd index with $|\alpha_{j^*}| \ge 1$. We evaluate it in the $K/2$ truncation, in which the formal density used for a K-atom ring has $K/2$ harmonic pairs. At $r_p = 1$ the loss points of the rule for $K = 16$, 20, 24, 28 and 30 are $\epsilon = 0.07171$, 0.05835, 0.04902, 0.04266 and 0.04080, so $K\epsilon = 1.15$, 1.17, 1.18, 1.19, 1.22: in a first lobe the count grows like $1/\epsilon$ (Fig. 3). To leading order in ϵ the coefficients are $c_{2k} = 2a_k(\epsilon/2)^k$, and a_k first changes sign at $k = 17$, which ends the first lobe. Past it the rule gives $K\epsilon^{1/2} = 6.29$, 6.31, 6.37, 6.45, 6.53 for $K = 34$, 36, 38, 40, 42, a local exponent $d\ln K / d\ln(1/\epsilon) = 0.61$; the 34- and 36-rings were solved, the larger ones were not. From $K = 38$ to 320 the rule's exponent is 0.59. If Conjecture 2 holds, the layout of the optimal ring alternates in blocks about 32 wide in K (15 to 17 even ring sizes each), so the vertex statement of [18] would fail on infinitely many intervals of ϵ accumulating at zero. This extrapolates the sign pattern of a_k beyond $k = 160$ and is not a measurement.

F. Toward the segment

As $r_m \to 0$ the ellipse collapses onto the segment $[-r_p, r_p]$, and both problems become the scalar channel with $A = r_p$; a scalar input with K_1 atoms is the ring with $2K_1 - 2$ atoms. Evaluated at $r_m = 0.005$ and extrapolated linearly in r_m, the loss rule with the linear-response correction places the first two scalar changes at $A = 1.6653$ and 2.9317, against 1.665925641 and 2.9075 (-0.04% and $+0.83\%$) [9], [17]; the copying condition alone gives 1.592 and about 3.2. The procedure was fixed before it was run, but the targets were known. From the third change on ($A \approx 4.0$) the formal density near the segment violates $|c_n| \le 2$, which every positive measure satisfies, and the rule is not defined there; the scalar count then follows the edge layers and the count law of Section III.

TABLE III

REGISTERED PREDICTIONS OF RING LOSSES AGAINST THE TRACKED OPTIMUM (CAPACITY). P1: FILLED REGION; P2: INPUT ON THE CURVE. MIXED: THE CURVE (5) WITH $\lambda = 0.02$. FORM S: THE LOSS CONDITION $b_K = c_K$ OF CONJECTURE 2; LR: WITH THE LINEAR-RESPONSE CORRECTION. KIND AND PLACE: X OR Y PAIR, SPLIT OR BIRTH, REGISTERED WITH THE VALUE AND OBSERVED. EACH VALUE WAS REGISTERED WITH AN ACCEPTANCE BAND, AND ALL FOURTEEN OBSERVED EVENTS FELL INSIDE THEIR BANDS (BANDS IN THE SUPPLEMENTARY LOG).

curve, r_p, problem, event	form: registered ϵ	observed ϵ	observed/predicted -1	kind, place
ellipse, 1, P1, 30-ring lost	S: 0.0408032574	(0.0408049, 0.0408055)	$+0.004$ to $+0.005\%$	X split
ellipse, 1, P1, 32-ring lost	S: 0.0399404809	(0.0399457, 0.0399463)	$+0.013$ to $+0.015\%$	Y birth
ellipse, 1, P1, 34-ring lost	S: 0.0342692971	(0.0342733, 0.0342734)	$+0.012\%$	Y split
ellipse, 1, P1, 36-ring lost	S: 0.0307639408	(0.0307656, 0.0307672)	$+0.005$ to $+0.011\%$	Y birth (not resolved)
mixed, 1.2, P1, 16-ring lost	S: 0.134287189	0.13432752	$+0.030\%$; LR 1×10^{-7}	X split
mixed, 1.2, P1, 18-ring lost	S: 0.12493239	0.1252793873	$+0.278\%$	X birth
mixed, 1.2, P1, 20-ring lost	S: 0.100908	0.1010178	$+0.11\%$; LR $\leq 7 \times 10^{-6}$	X split
mixed, 1.2, P1, 22-ring lost	S: 0.087130	(0.0872266, 0.0872344)	$+0.11$ to $+0.12\%$	X birth
mixed, 1.2, P1, 24-ring lost	S: 0.074900	(0.07495, 0.07500)	$+0.07$ to $+0.13\%$	X split
mixed, 1.2, P1, 26-ring lost	S: 0.065283	(0.0653262, 0.0653320)	$+0.07$ to $+0.08\%$	X birth
ellipse, 3, P1, 18-ring lost	LR: 0.9518201483	0.9519100801	$+0.0094\%$	Y birth
ellipse, 3, P2, 20-ring lost	LR: 0.9046658875	0.9047303949	$+0.0071\%$	Y split
ellipse, 3, P2, 22-ring lost	LR: 0.8625141199	0.8625613848	$+0.0055\%$	Y birth
ellipse, 3, P2, 24-ring lost	LR: 0.8242977666	0.8243330	$+0.0043\%$	Y split

On the ellipse $r_p = 3$ the form S alone missed the 18-ring by $+1.18\%$, after misses of $+8.1$, $+4.8$, $+3.3$, $+2.3$, $+1.8$ and $+1.4\%$ at $K = 6, \ldots, 16$. The mixed curve at $r_p = 1.2$ was checked to satisfy P1 on an interior grid.

TABLE IV

RING LOSSES REGISTERED IN ADVANCE IN THE OTHER TWO PROBLEMS, ELLIPSE $r_p = 1$, AGAINST THE TRACKED OPTIMUM. NPMLE: GAUSSIAN SOURCE, $s = 2$, ϵ INCREASING. DA: DETERMINISTIC ANNEALING AT $r_m = 0.7$, $s = 2$, TEMPERATURE DECREASING. ENTRIES ARE OBSERVED/PREDICTED -1; ALL REGISTERED BANDS WERE $\pm 1\%$ (RULE) AND $\pm 0.02\%$ (LR).

problem, K lost	kind, place	rule alone	LR
NPMLE, 14	Y birth	$+0.217\%$	$\leq 3.5 \times 10^{-6}$
NPMLE, 16	X split	$+0.019\%$	$\leq 3 \times 10^{-6}$
NPMLE, 18	X birth[a]	$+0.61\%$	–
LFP, 14	Y birth	$+0.076\%$	1.2×10^{-8}
LFP, 16	Y split	$+0.057\%$	$\leq 5 \times 10^{-7}$
DA, 8	Y split	$+0.37\%$	–
DA, 10	Y birth	$+0.36\%$	–
DA, 12	Y split	$+0.18\%$	–
DA, 14	Y birth	$+0.31\%$	–
DA, 16	Y split	$+0.007$ to $+0.039\%$	–

[a] Observed; the kind of this event was not registered.

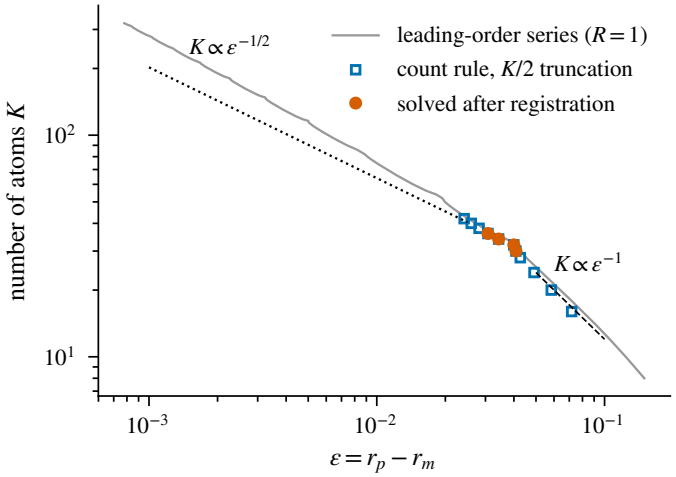


Fig. 3. The number of atoms grows like $1/\epsilon$ in a first lobe and more slowly beyond it. Ellipse $r_p = 1$: the count rule in the $K/2$ truncation at $R = r_p - \epsilon/2$ (squares), the leading-order series at $R = 1$ (line), optima solved after their prediction was registered (dots; rows 1–4 of Table III), and the guides $K = 1.2/\epsilon$ (dashed) and $K = 6.4\,\epsilon^{-1/2}$ (dotted).

VII. A RING WITHOUT VERTEX ATOMS, BY A COMPUTER-ASSISTED PROOF

A. The result

Dytso, Barletta and Kramer [18] study the channel $Y = HX + Z$ with a real 2×2 matrix H, Z standard normal in $\mathbb{R}^2$ and the peak constraint $\|X\| \leq 1$, through the equivalent problem of maximizing $I(X; X + Z)$ over inputs in the filled ellipse with semi-axes $r_p \geq r_m$ [18, eqs. (1)–(3)]. Their Theorem 4 reads, in our notation: if $0 \leq r_m \leq r_p$, then both ends of the major axis, $(\pm r_p, 0)$, lie in the support of the capacity-achieving input. Our problem P1 with unit noise is this problem. The copying rule predicted rings without these atoms on the ellipse $r_p = 1$, for ϵ between 0.03077 and 0.0408049, where the optimal rings have 32, 34 and 36 atoms. We certify one of them.

Theorem 3 (computer-assisted)**.** *For $r_p = 1$ and $r_m = 0.9597$, the capacity-achieving input of the channel $Y = X + Z$ with X in the filled ellipse is a symmetric ring of 32 atoms whose angles and masses lie within 6.82×10^{-66} of the midpoints of the certified box. Table V prints these midpoints rounded to 20 decimals, and the supplementary material gives them in full. In particular $(\pm r_p, 0)$ is not in its support. The capacity is $C = 0.39064923014816081039\ldots$ nats (0.563588 bits), enclosed to 7×10^{-74}.*

The proof is given in Section VII-B; the bound on the quadrature error is proved in Appendix E. The capacity-achieving input is unique, since the output law determines the input law (the Gaussian characteristic function has no zeros) and the output entropy is strictly concave in the output law. The channel $Y = HX + Z$ with $H = \mathrm{diag}(1, 0.9597)$ and $\|X\| \leq 1$ realizes this input. Theorems 1–3 of [18] are not affected, since their proofs do not use Theorem 4. The two atoms nearest the end $(r_p, 0)$ sit at $t = \pm 2.5864°$ (Fig. 4). At

TABLE V
THE CERTIFIED 32-ATOM RING AT $r_p = 1$, $r_m = 0.9597$: ONE REPRESENTATIVE PER ORBIT, MASS PER ATOM. MIDPOINTS OF THE CERTIFIED BOX ROUNDED TO 20 DECIMALS; THE BOX RADIUS IS AT MOST 6.82×10^{-66}.

orbit	parameter angle t_j (rad)	mass per atom
G1	0.04514041824277957604	0.02587729920137166890
G2	0.23892408883896762725	0.03897966931781938891
G3	0.45323176005362528202	0.03545268828046724276
G4	0.66308206341144781172	0.03249339856300210627
G5	0.86870324583193416037	0.03056684647659560933
G6	1.07128670703876574422	0.02940210562835088532
G7	1.27186053563821704426	0.02875615919397271245
G8	1.47128043156772716194	0.02847183333842038607

Atoms at $(r_p \cos t, r_m \sin t)$, $t = \pm t_j, \pi \pm t_j$.

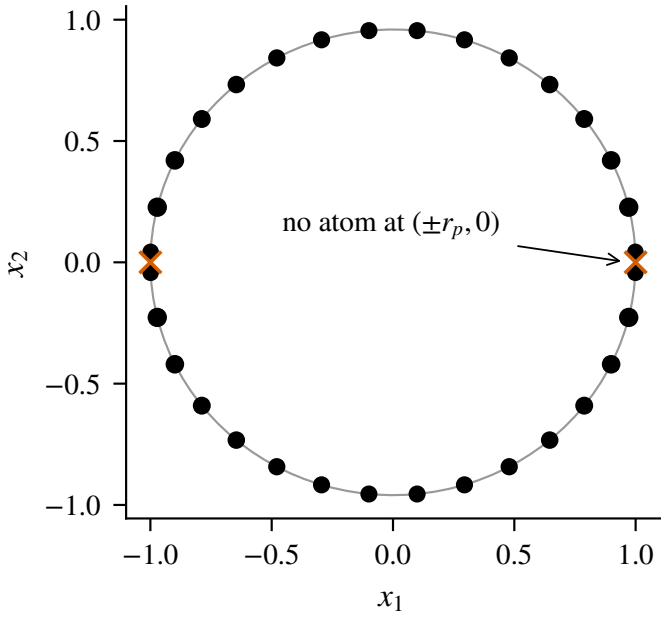


Fig. 4. The certified optimum carries no atom at the ends of the major axis. Ellipse $r_p = 1$, $r_m = 0.9597$; marker area is proportional to mass. The pair next to each end, at $t = \pm 2.5864°$, carries the smallest mass, 0.0259.

the end itself the certificate gives $D_\pi - C = -1.1276 \times 10^{-50}$, and at the end of the minor axis -1.4177×10^{-48}, so neither is a support point.

B. The certificate

All arithmetic is midpoint-radius ball arithmetic (python-flint 0.9.0, the interface to the Fast Library for Number Theory (FLINT) and Arb [33]). The unknowns are the eight angles, the eight masses and C; the equations are $D_\pi = C$ and $\partial_t D_\pi = 0$ at one atom of each orbit, and $4\sum_j m_j = 1$. The proof has four steps, with every constant in Appendix E.

1) *Quadrature with an error bound.* The expectation in D_π is replaced by a trapezoid sum on a square grid, with an error bound from analyticity in a strip [34] (Lemma 7).
2) *An exact solution.* A Krawczyk test [35]–[37] shows that a box of radius 6.82×10^{-66} around the computed ring contains exactly one solution of the optimality equations, with every mass above 0.025877; the capacity C, one of the unknowns, carries its own smaller radius in the same box.
3) *The boundary.* On 230 pieces of $t \in [0, \pi/2]$, Taylor enclosures with Cauchy remainders show $D_\pi - C \le -1.74 \times 10^{-53}$ away from the atoms and $\partial_t^2 (D_\pi - C) < 0$ in a zone around each atom, uniformly over the box.
4) *The interior.* The largest atom radius is $0.99992 < 1$, so by the Hessian identity of Section II D_π is strictly convex on $\mathbb{R}^2$, and every interior point lies strictly below the boundary maximum.

Together with the sufficiency of the optimality condition [2], [21], these steps prove Theorem 3. The proof of Lemma 7, which bounds the quadrature error, is in Appendix E. A separate evaluator with no shared code (a trapezoid rule at 256 bits) reproduces $D_\pi - C$ below 10^{-76} at the atoms and the two vertex values to the digits shown. A second rigorous check shares nothing with the ring. Putting mass $f(t_l)/128$ on 128 equally spaced points of the curve, with f the formal density truncated at four harmonic pairs ($\min f = 0.899$), gives an admissible input whose mutual information bounds C from below, and its output law q bounds it from above through $C \le \sup_x D(P_{Y|x} \| q)$ [26]. The four coefficients of f are fixed by the definition of Section II and printed by the check script of the supplementary material. With the same machinery this gives $0.3906492301481604 \le C \le 0.3906492301483361$ (ends rounded outward), which contains the certified value, and the two bounds are 1.8×10^{-13} apart. Before the outward rounding the lower end is 0.39064923014816047684, with a radius below its last digit. By the certified value of C, the mutual information of this 128-point input is within 3.4×10^{-16} nats of capacity. The structure of the exact ring matters for the proof and for the correction of [18], and it matters little for a design.

C. Where the published proof breaks

The proof of Theorem 4 in [18] lets $t < r_p$ be the largest abscissa in the support and uses their Lemma 7, $\partial D/\partial x_1 \ge x_1 - t > 0$ for $x_1 > t$, to conclude that $D(x_1, x_2) > D(t, x_2)$ for $x_1 > t$, which, they write, would violate the Kuhn–Tucker condition. The implicit step is that some (t, x_2) with feasible points to its right is a support point. On the certified ring the largest abscissa is $t = \cos(2.5864°) = 0.998981$ (here t is an abscissa, as in [18], and not the curve parameter of Section VI, and the angle is in degrees, whereas Table V gives angles in radians), attained by the pair at $x_2 = \pm 0.043307$. On those two lines the ellipse ends at $x_1 = t$, so no feasible point lies to the right of the support points, and the comparison yields no violation. On the line $x_2 = 0$, where feasible points with $x_1 \in (t, r_p]$ exist, $(t, 0)$ is not a support point. What the argument does prove is the following.

Proposition 2. *Let $0 < r_m \le r_p$ and let t be the largest abscissa of a support point. Then no feasible point lies to the right of any support point with abscissa t; that is, every such point (t, x_2) satisfies $t = r_p(1 - x_2^2/r_m^2)^{1/2}$.*

Proof. At a support point $D(t, x_2) = C$. Differentiating under the integral and using $\nabla \log p_\pi(y) = \delta_\pi(y) - y$ (Section II)

gives $\nabla D(x) = x - \mathbb{E}\,\delta_\pi(x+Z)$. The first coordinate of the Bayes rule δ_π is at most t, so $\partial D/\partial x_1 \geq x_1 - t > 0$ for $x_1 > t$, and $D(x_1, x_2) > C$ for $x_1 > t$ (Lemma 7 of [18]), which the optimality condition forbids at feasible points. □

So the support points of maximal abscissa lie on the boundary, and the proposition does not decide whether $(r_p, 0)$ is one of them. For $r_p \leq \sqrt{2}$, where every support point lies on the boundary, the argument gives nothing more.

VIII. Leaving the boundary: births inside a filled region

A. The center test

On a filled region the boundary ring stays optimal while its optimality function also stays below the constant inside. On the grids we checked, the largest interior value sat at the center whenever it came close to the constant, so the boundary optimum is lost to an atom born at the center, the counterpart of the birth at the origin on an interval. The formal density predicts when this happens. Harmonic j of the input is suppressed near the origin by the factor of Section VI-C, so the value $F(0)$ depends mainly on the low harmonics, and those are the ones the ring copies. We therefore evaluate the optimality function of the formal density at the center, $F_f(0)$, and subtract C, taken as the mean of F_f over the curve; the difference is the center excess. On a disc $f = 1$ exactly, and the excess vanishes at radius 2.4534 for capacity (2.454 in Table I of [9]) and at 1.53499 for the LFP (1.535 in the same table). In that table the LFP value is the largest radius at which the uniform law on the circle maximizes the minimum mean-square error. Berry [38] gives the same value, 1.53499, as the radius below which the uniform law on the circle is least favorable. For the NPMLE with a Gaussian source of standard deviation $s = 2$ the excess vanishes at 1.8982.

B. Registered center births

Table VI lists the registered births and Fig. 5 the curves they lie on. The boundary rings were solved afterwards in 128- to 160-bit arithmetic, and their center excess was read at several values of r_m; its zero is the observed birth. Along the curve their optimality function stays within 2×10^{-9} of the constant. For the center excess, the number of atoms in the ring did not need to be resolved. The center excess was the same for LFP rings of 16, 18 and 20 atoms to 10^{-13} on the ellipse, and for NPMLE rings of 16 and 20 atoms to 1.2×10^{-11}. Below about 14 atoms it still depends on the ring size. The three births on the ellipses passed, within 3×10^{-7} in r_m. On the mixed curves of Table VI the boundary rings that the solver first returned have negative masses, so they are not probability laws. Followed from the ellipse by continuation in λ with every mass kept positive, the NPMLE ring loses its pair at the ends of the major axis near $r_m = 1.7393$, and the ring of 18 atoms that follows passes the registered test, with the zero at 1.7363463. The LFP ring, followed the same way at $r_m = 1.4036$, first merges an orbit into a pair at the ends of the minor axis near $\lambda = 0.0605$ and then loses its pair at the ends of the major axis near $\lambda = 0.0923$. The ring of 14 atoms that follows passes the registered test, with the zero at 1.4015695. For capacity, the birth on the mixed curve is not confirmed. Since the ring copies the low harmonics of the density, this tests a consequence of copying for an interior quantity; it is not independent evidence for Conjecture 1.

TABLE VI

Registered center births: the r_m at which the center excess of the boundary ring vanishes. Mixed λ: the curve (5) with that λ. NPMLE: Gaussian source, $s = 2$; [b] rings at 200 Gauss–Hermite nodes; the ellipse value is 1.741648 at 120 nodes and 1.7416352 at 300 nodes; [c] not confirmed: the boundary rings found have negative masses. [d] Ring of 18 atoms, after the pair at the ends of the major axis leaves the support near $r_m = 1.7393$. [e] Ring of 14 atoms, after an orbit merges at the ends of the minor axis and the pair at the ends of the major axis leaves the support. The LFP on the mixed curve uses $\lambda = 0.1$, where its formal density converges.

problem	region, r_p	registered (band)	observed
capacity	ellipse, 2.8	2.1874219 ($\pm 10^{-5}$)	2.18742185
LFP	ellipse, 1.7	1.4038104 ($\pm 2 \times 10^{-4}$)	1.4038104
NPMLE	ellipse, 2.1	1.7416352 ($\pm 2 \times 10^{-4}$)	1.741635[b]
capacity	mixed 0.2, 2.8	2.1756059 ($\pm 2 \times 10^{-4}$)	–[c]
LFP	mixed 0.1, 1.7	1.4015695 ($\pm 2 \times 10^{-4}$)	1.4015695[e]
NPMLE	mixed 0.2, 2.1	1.7363460 ($\pm 2 \times 10^{-4}$)	1.7363463[d]

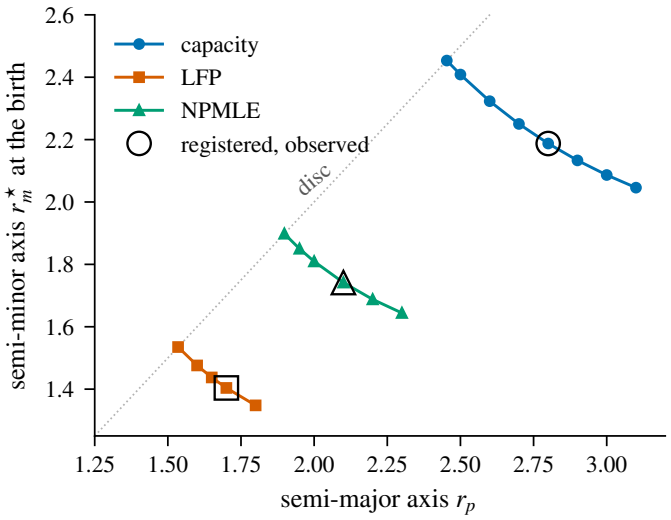


Fig. 5. The formal density predicts where the optimum on a filled ellipse first needs an atom at the center, in all three problems. Lines: predicted $r_m^\star(r_p)$ for capacity, the LFP and the NPMLE ($s = 2$); the first point of each curve is the disc. Open symbols: registered births observed on boundary rings (Table VI), within 3×10^{-7} of the prediction. Each curve ends where the formal density stops converging.

For the 2×2 MIMO channel this suggests where the optimum stays on the boundary. Along the capacity curve the birth lies at $r_m^\star = 2.40857, 2.32318, 2.25017, 2.18742, 2.13331$ and 2.08650 for $r_p = 2.5, 2.6, \ldots, 3.0$. Below the circle value the center excess is negative wherever we evaluated a converged density, and the solved rings at $r_p = 3$ approach the small negative scalar value as $r_m \to 0$. We therefore conjecture, from these grids only, that the optimal input stays on the boundary ellipse for every r_m when $r_p \leq 2.4534$, beyond the range $r_p \leq \sqrt{2}$ of [18, Thm. 1], and for larger r_p while $r_m \leq r_m^\star(r_p)$. Nothing between $\sqrt{2}$ and 2.4534 was checked on an interior grid, so the conjecture for that range of r_p is an

TABLE VII
REGISTERED TESTS THAT DID NOT PASS AND BEAR ON THE RESULTS OF SECTIONS IV-B, VI AND VIII

outcome	log entry
failed: form S on the ellipse $r_p = 3$, 18-ring ($+1.18\%$)	1356
failed: extremal form, mixed curve, 18-ring (-2.81%)	1331
failed: inner pairs at the largest ϵ_i (three problems)	1438–1477
failed: first births law, residual step at $K_b = 71$	1820
failed: first births law, residual ratio at $K_b = 78$	1916
missed: table of the first variant, NPMLE transition $K_b = 93$	2040
failed: table of the first variant, capacity birth $K_b = 94$	2160
not confirmed: center birth on the mixed curve for capacity (rings with negative masses)	1452

extrapolation of the disc value and of the grids at larger r_p, and the limit of thin ellipses needs an argument.

IX. WHAT DOES NOT CARRY OVER, AND WHAT IS OPEN

a) What does not carry over: Some structure holds only in part. The rule for rings was tested near circles, on ellipses up to $r_p = 3$ and on one other family of curves. The inner rings that appear after a birth follow the rule to first order in their eccentricity only, and the second-order term differs between the problems. The ring counts of the LFP on a disc differ from those of the capacity and the NPMLE by about half an event spacing; the Supplementary Notes give these comparisons and the full list of tests that failed. Table VII lists the registered tests that did not pass and bear on the results above. The entry numbers are those of the public registration log.

b) What is open: The statements below are not proved here, and we do not know whether they are true in general.

1) *Monotonicity of $K(A)$.* By Section V-A it reduces to two statements about the innermost atoms. The missing step is the inequality $\Sigma_+ > \Sigma_-$ of Section V-A for the response of the edge region to the motion of the wall. The margin $\Sigma_+/\Sigma_- - 1$ is positive in every stored state; it is 5 to 9 percent for $K = 19$ to 81 and 4.7 percent at the smallest of the rows $K = 83$ to 130. An extrapolation puts its limit near 3.4 percent.
2) *The upper bound $K = O(A^{4/3})$.* Only $O(A^2)$ is known [12], and the measured law says order $A^{4/3}$. We have no route to a proof.
3) *The theorem behind the law.* The edge constants and the exponentially small corrections of the births law are measured and not derived (for the capacity strip β only two-sided bounds are proved, Section III), and the phase condition behind the bulk law is not rigorous. The aliasing terms of the derivation of the series are not bounded.
4) *Copying and loss.* Conjectures 1 and 2 of Section VI are tested and not proved.
5) *Total positivity of the whole matrix X.* We prove the Gram part, and the whole matrix at the 91 optimal states of Proposition 1. A proof for all K needs $t^* > 1$ (Section V-B), and we have no bound for it.

APPENDIX A
PROOF OF THEOREM 1

Conventions. The functions φ and Φ are the standard normal density and distribution function, $\bar\Phi = 1 - \Phi$, φ_s is the $\mathcal{N}(0, s^2)$ density, $h_1 = \frac{1}{2}\log(2\pi e)$, and $\hat f(\xi) = \int e^{-i\xi y} f(y)\,dy$. The logarithmic mean of 1 and $r > 0$ is $L(1, r) = (r - 1)/\log r = \int_0^1 r^s\,ds$ with $L(1, 1) = 1$; it is increasing in r. Let P be the capacity-achieving input on $[-A, A]$. By Smith [2] it is unique and has finitely many atoms. It is symmetric: the channel and the constraint are invariant under $x \mapsto -x$, so the mirror image of P is also optimal, and the optimum is unique. Write $C = C(A)$ for the capacity in nats, $C' = C + h_1$, $c^* = e^{-C'}$, $p = \varphi * P$ for the output density and $g = \log(p/c^*)$. Put $\lambda_1 = 1.1648$, $\ell = e^{-\lambda_1}$, $L_\ell = L(1, \ell) = (1 - \ell)/\lambda_1$ and $\rho_1 = (e - 1)/L_\ell$.

A. Reference level and bounds with absolute constants

Lemma 2. *For every $A > 0$ we have $e^{-1/2}/(2A + \sqrt{2\pi}) \le c^* \le 1/(2A)$, and the Kuhn–Tucker conditions of the optimum read*

$$(\varphi{*}g)(x) \ge 0 \text{ for } |x| \le A, \qquad (\varphi{*}g)(x) = 0 \text{ for } x \in \operatorname{supp} P. \tag{7}$$

Proof. Let U be uniform on $[-A, A]$. Then $C \ge I(U; U + Z) = h(U + Z) - h(Z) \ge h(U) - h(Z)$ (conditioning on Z does not increase the entropy of $U + Z$), so $C' \ge \log 2A$ and $c^* \le 1/(2A)$. For the other bound let $q(y) = e^{-(|y|-A)_+^2/2}/(2A + \sqrt{2\pi})$, a probability density. For every input law, $I(X; Y) = \mathbb{E}\, D(\mathcal{N}(X, 1)\|q) - D(p\|q) \le \max_{|x|\le A} D(\mathcal{N}(x, 1)\|q)$, and for $|x| \le A$ we have $D(\mathcal{N}(x, 1)\|q) = -h_1 + \log(2A + \sqrt{2\pi}) + \frac{1}{2}\mathbb{E}(|x + Z| - A)_+^2 \le -h_1 + \log(2A + \sqrt{2\pi}) + \frac{1}{2}$, because $(|x + Z| - A)_+ \le |Z|$. Hence $C' \le \log(2A + \sqrt{2\pi}) + \frac{1}{2}$. The Kuhn–Tucker conditions [2] say $D(\mathcal{N}(x, 1)\,\|\,p) \le C$ for $|x| \le A$, with equality on $\operatorname{supp} P$. Since $D(\mathcal{N}(x, 1)\|p) = -h_1 - \int \varphi(y - x)\log p(y)\,dy$, they read $\int \varphi(y - x)(\log p(y) + C')\,dy \ge 0$ with equality on the support, which is (7). □

Lemma 3. *For every $A > 0$, with $P_{\mathrm{loc}} = e/\varphi(\frac{1}{2})$, the sum $S_1 = \sum_{k\ge 0} \max_{s\in[k,k+1]} s\varphi(s)$ and $\mu_B = S_1 P_{\mathrm{loc}}/\ell$:*

(a) *$p \le e\,c^*$ on $\mathbb{R}$;*
(b) *$P([y - \frac{1}{2}, y + \frac{1}{2}]) \le P_{\mathrm{loc}}\,c^*$ for every real y;*
(c) *$p > \ell\,c^*$ on $[-A, A]$;*
(d) *$\ell c^* e^{-t^2/2-\mu_B t} \le p(\pm(A + t)) \le e\,c^* e^{-t^2/2}$ for $t \ge 0$;*
(e) *$|p'| \le L_{\mathrm{e}}\,c^*$ on $\mathbb{R}$, with $L_{\mathrm{e}} = 2S_1 P_{\mathrm{loc}}$.*

In particular $g \le 1$ on $\mathbb{R}$, while $g > -\lambda_1$ on $[-A, A]$ and $|g(\pm(A + t))| \le \lambda_1 + t^2/2 + \mu_B t$ (where $g \ge 0$, this follows from $g \le 1 < \lambda_1$).

Proof. The function $\psi(y) = \log p(y) + y^2/2 = \log \sum_j w_j(2\pi)^{-1/2}\exp(x_j y - x_j^2/2)$ is a logarithm of a sum of exponentials of affine functions, hence convex; so $(\log p)'' \ge -1$.

(a) Let y_0 be a global maximizer of p and $M_A = p(y_0)$. If every atom x_j had $|x_j - y_0| > 1$, then $p''(y_0) = \sum_j w_j((x_j - y_0)^2 - 1)\varphi(y_0 - x_j) > 0$, contradicting maximality. So some atom x has $|x - y_0| \le 1$. Since $(\log p)'(y_0) = 0$ and $(\log p)'' \ge$

-1, Taylor's formula gives $\log p(y) \ge \log M_A - (y-y_0)^2/2$. By (7) at the atom x we have $-C' = \int \varphi(y-x)\log p(y)\,dy \ge \log M_A - \frac12(1+(x-y_0)^2) \ge \log M_A - 1$. Hence $M_A \le e\,c^*$.

(b) If $|\theta - y| \le \frac12$ then $\varphi(y-\theta) \ge \varphi(\frac12)$, so $\varphi(\frac12)P([y-\frac12, y+\frac12]) \le p(y) \le e\,c^*$.

(c) Fix $x_0 \in [-A, A]$ and put $\lambda = -g(x_0)$; if $\lambda \le 0$ there is nothing to show, so let $\lambda > 0$. Let $R' > 0$ and $0 \le u \le R'$. Convexity of ψ on $[x_0, x_0 + R']$ and $g = \psi - y^2/2 - \log c^*$ give

$$g(x_0+u) \le (1-\tfrac{u}{R'})g(x_0) + \tfrac{u}{R'}g(x_0+R') + \tfrac12 u(R'-u),$$

the last term being the chord defect of $y^2/2$. With $g(x_0) = -\lambda$ and $g \le 1$ from (a), this is $g(x_0+u) \le -\lambda - \frac{u^2}{2} + u\big(\frac{1+\lambda}{R'} + \frac{R'}{2}\big)$, and the same holds for $g(x_0-u)$ by using $[x_0 - R', x_0]$. Minimizing over $R' \ge u$ gives $g(x_0 \pm u) \le G_\lambda(u)$, where $G_\lambda(u) = -\lambda + R_\lambda u - u^2/2$ for $u \le R_\lambda := \sqrt{2(1+\lambda)}$ and $G_\lambda(u) = 1$ for $u > R_\lambda$. Each expression in the infimum equals $-(1-u/R')\lambda + u/R' + \frac12 u(R'-u)$ and is therefore nonincreasing in λ; so $G_\lambda(u)$ is nonincreasing in λ. By (7),

$$\begin{aligned} 0 \le (\varphi * g)(x_0) &= \int \varphi(u)\, g(x_0+u)\,du \\ &\le J(\lambda) := 2\int_0^\infty \varphi(u) G_\lambda(u)\,du, \end{aligned}$$

and J is nonincreasing. Writing $R = R_\lambda$, $j_a = \Phi(R) - \frac12$, $j_b = \varphi(0) - \varphi(R)$ and $j_c = j_a - R\varphi(R)$ (the integrals of φ, $u\varphi$ and $u^2\varphi$ over $[0,R]$), we get the closed form $J(\lambda) = 2[-\lambda j_a + R j_b - \frac12 j_c] + 2\bar\Phi(R)$. In ball arithmetic with 256 bits, $J(1.1648) = -5.001074\cdot 10^{-5}$ to seven digits, with a ball radius below 10^{-75}, and $J(1.1647) > 0$ (the evaluation is a short script in the supplementary repository). If $\lambda \ge \lambda_1$ then $J(\lambda) \le J(\lambda_1) < 0$, a contradiction. Hence $\lambda < \lambda_1$.

(d) For $\theta \in [-A, A]$ we have $(A+t-\theta)^2 \ge (A-\theta)^2 + t^2$, so $p(A+t) \le e^{-t^2/2}p(A) \le e\,c^* e^{-t^2/2}$ by (a). Next, $(A+t-\theta)^2 = (A-\theta)^2 + 2t(A-\theta) + t^2$ gives $p(A+t) = e^{-t^2/2}p(A)\,\mathbb{E}[e^{-t(A-X)} \mid Y = A] \ge e^{-t^2/2}p(A)e^{-t\mu_A}$ by Jensen, with $\mu_A = \mathbb{E}[A - X \mid Y = A]$. Covering $[-A, A]$ by the intervals $I_k = [A-k-1, A-k]$ and using (b), $p(A)\mu_A = \int (A-\theta)\varphi(A-\theta)\,dP \le S_1 P_{\rm loc} c^*$; with $p(A) > \ell c^*$ from (c) this gives $\mu_A \le \mu_B$. The statements at $-A-t$ follow by symmetry.

(e) $p'(y) = \int (\theta - y)\varphi(y-\theta)\,dP(\theta)$. Cover the line by the unit intervals $y \pm [k, k+1]$ and use (b) on each. □

Lemma 4. *On $[-A, A]$: $(p - c^*)g \ge c^* L_\ell\, g^2$ and $(p-c^*)^2 \le (e-1)\,c^*(p-c^*)g$.*

Proof. Put $r = p/c^* \in (\ell, e]$ (Lemma 3(a),(c)). Then $(r-1)\log r = L(1,r)(\log r)^2$ and $(r-1)^2 = L(1,r)(r-1)\log r$, and $L_\ell \le L(1,r) \le L(1,e) = e - 1$. □

B. A weighted identity and the decay of $p - c^*$

Let $\mathcal{M} = c^*\mathbf{1}_{[-A,A]}\,dx - P$ (a signed measure of mass $2Ac^* - 1 \le 0$), $F = \varphi * \mathcal{M}$ and $w(x) = \bar\Phi(A-x) + \bar\Phi(A+x)$. Since $\varphi * \mathbf{1}_{[-A,A]} = 1 - w$, we have $F = -(p-c^*) - c^* w$ on $\mathbb{R}$. For a bounded weight $\chi \ge 0$ on $[-A, A]$ put $E_2 = \int_{-A}^{A} \chi\,(p - c^*)\,g\,dx$, which is nonnegative because $(p-c^*)g \ge 0$, and $C_\chi = \varphi * (\chi\mathcal{M}) - \chi F$ on $[-A, A]$.

Proposition 3. *The weighted fitting error satisfies $E_2 \le c^*\big|\int_{-A}^{A} \chi g w\big| + \big|\int_{|y|>A} g\,\varphi * (\chi\mathcal{M})\big| + \big|\int_{-A}^{A} g C_\chi\big|$.*

Proof. Let $\Omega = c^*\int_{-A}^{A}\chi(C' - \mathcal{T})\,dx$ with $\mathcal{T}(x) = -\int\varphi(y-x)\log p(y)\,dy$, so that $C' - \mathcal{T} = \varphi * g$. By (7), $\Omega \ge 0$ and $\int \chi(\varphi * g)\,dP = 0$; hence $\Omega = \int \chi(\varphi * g)\,d\mathcal{M} = \int g\,\varphi * (\chi\mathcal{M})\,dy$. (Fubini applies: Lemma 3 gives $|g(y)| \le \lambda_1 + (|y| - A)_+^2/2 + \mu_B(|y|-A)_+$, and $(|x+z|-A)_+ \le |z|$ for $|x| \le A$.) Split $\int_{\mathbb{R}}$ into $[-A, A]$ and $|y| > A$, write $\varphi * (\chi\mathcal{M}) = \chi F + C_\chi$ on $[-A, A]$ and $g\chi F = -\chi(p-c^*)g - c^*\chi g w$. This gives $\Omega = -E_2 - c^*\int \chi g w + \int_{|y|>A} g\,\varphi * (\chi\mathcal{M}) + \int g C_\chi$, and $\Omega \ge 0$ gives the claim. □

Proposition 4. *Let $0 < c_0 < c_{0,\max} := (4\rho_1)^{-1/3}$, and put $\Lambda_0 = c_0 A^{2/3}$ and $\chi(x) = \exp(\Lambda_0(1 - x^2/A^2))$. There are A_0 and $\bar E < \infty$, depending only on c_0, such that $E_2 \le \bar E\,c^*$ for $A \ge A_0$.*

Proof. Let $\gamma = \Lambda_0/(2A^2)$, so $\chi^{1/2}(x) = e^{\gamma(A^2 - x^2)}$, and take A so large that $\Lambda_0 \le A/2$; then $\gamma \le 1/(4A) < \frac12$. Put $t_\pm = 1/(1 \pm 2\gamma)$, $\Delta = t_- - t_+ = 4\gamma/(1-4\gamma^2)$, $\mathcal{N} = \chi^{1/2}\mathcal{M}$ and $N = 2e^{\Lambda_0/2} \ge \|\mathcal{N}\|_{TV}$ (since $\|\mathcal{M}\|_{TV} \le 2$). Let $H_t = \varphi_{\sqrt t} * \mathcal{N}$, so that $\hat H_t(\xi) = e^{-t\xi^2/2}\hat{\mathcal{N}}(\xi)$ and $|\hat{\mathcal{N}}| \le N$. Let $\mathsf{P} = \chi^{1/2}F$ and $\mathsf{Q} = \chi^{-1/2}\,\varphi * (\chi\mathcal{M})$ on $\mathbb{R}$; then $C_\chi = \chi^{1/2}(\mathsf{Q} - \mathsf{P})$ on $[-A, A]$.

Step 1 (Gaussian formulas). Completing the square, $\gamma(x^2 - \theta^2) - \frac12(x-\theta)^2 = -\frac{1}{2t_+}(\theta - t_+ x)^2 + \frac{2\gamma^2 x^2}{1+2\gamma}$. Hence $\mathsf{Q}(x) = \int e^{\gamma(x^2-\theta^2)}\varphi(x-\theta)\,d\mathcal{N}(\theta) = \alpha_+(x) H_{t_+}(t_+ x)$, and in the same way, with γ replaced by $-\gamma$ (allowed because $2\gamma < 1$), $\mathsf{P}(x) = \alpha_-(x) H_{t_-}(t_- x)$. Here $\alpha_\pm(x) = (1 \pm 2\gamma)^{-1/2}\exp(2\gamma^2 x^2/(1\pm 2\gamma))$, and $\alpha_- \ge 1$.

Step 2 (three terms). The difference is $\mathsf{Q} - \mathsf{P} = (\alpha_+ - \alpha_-)H_{t_+}(t_+x) + \alpha_-[H_{t_+} - H_{t_-}](t_+x) + \alpha_-[H_{t_-}(t_+x) - H_{t_-}(t_-x)]$. Let $A_1 = \sup_{[-A,A]}|\alpha_+ - \alpha_-|$ and $A_2 = \sup_{[-A,A]}\alpha_-$; both are attained at $|x| = A$, because α_- and $\alpha_- - \alpha_+$ increase with x^2: $(1-2\gamma)^{-1/2} \ge (1+2\gamma)^{-1/2}$ and $2\gamma^2/(1-2\gamma) \ge 2\gamma^2/(1+2\gamma)$. In $L^2[-A, A]$, using $\int H(t_+x)^2 dx \le t_+^{-1}\|H\|^2$, the three terms are at most $A_1 t_+^{-1/2}\|H_{t_+}\|$, $A_2 t_+^{-1/2}\|H_{t_+} - H_{t_-}\|$ and $A_2 \Delta A\, t_+^{-1/2}\|H'_{t_-}\|$; the last one because $H(t_-x) - H(t_+x) = \int_{t_+}^{t_-} xH'(ux)\,du$, so by Cauchy–Schwarz its square integral over $[-A, A]$ is at most $A^2\Delta^2\|H'\|^2/t_+$.

Step 3 (Plancherel with one band B). Let $h = \|H_{t_-}\|$ and $B > 0$, and split each Plancherel integral at $|\xi| = B$, using $e^{-t_+\xi^2} = e^{\Delta\xi^2}e^{-t_-\xi^2}$, the bound $|\hat{\mathcal{N}}| \le N$ and $|e^{-t_+\xi^2/2} - e^{-t_-\xi^2/2}| \le e^{-t_+\xi^2/2}\Delta\xi^2/2$. With $R_k(t, B) = \int_{|\xi|>B}\xi^{2k}e^{-t\xi^2}d\xi/(2\pi)$,

$$\begin{aligned} \|H'_{t_-}\|^2 &\le B^2 h^2 + N^2 R_1(t_-, B), \\ \|H_{t_+} - H_{t_-}\|^2 &\le (e^{\Delta B^2/2} - 1)^2 h^2 + N^2(\tfrac{\Delta}{2})^2 R_2(t_+, B), \\ \|H_{t_+}\|^2 &\le e^{\Delta B^2} h^2 + N^2 R_0(t_+, B). \end{aligned}$$

Hence $\|\mathsf{Q} - \mathsf{P}\|_{[-A,A]} \le \delta_1 h + \rho_N$ with $\delta_1 = t_+^{-1/2}[A_1 e^{\Delta B^2/2} + A_2(e^{\Delta B^2/2} - 1) + A_2\Delta A B]$ and $\rho_N = t_+^{-1/2}N[A_1 R_0^{1/2} + A_2\frac{\Delta}{2}R_2^{1/2} + A_2\Delta A R_1^{1/2}]$, where $R_0 = R_0(t_+, B)$, $R_2 = R_2(t_+, B)$ and $R_1 = R_1(t_-, B)$.

Step 4 (back to P*).* By Step 1 and $\alpha_- \ge 1$ we have $h^2 \le \int \mathsf{P}(y/t_-)^2 dy = t_- \|\mathsf{P}\|^2$. On $[-A,A]$ we have $\mathsf{P} = -\chi^{1/2}(p-c^*) - c^*\chi^{1/2} w$, and $|\mathsf{P}| \le |F|$ outside because $\chi \le 1$ there. By Lemma 4, $\int \chi (p-c^*)^2 \le (e-1)c^* E_2$. With $\tau_w^2 = \int_{-A}^{A} \chi w^2$ and $\|F\|_{\mathrm{out}} = \|F\|_{L^2(|x|>A)}$ this gives $h \le t_-^{1/2}[((e-1)c^*E_2)^{1/2} + c^*\tau_w + \|F\|_{\mathrm{out}}]$.

Step 5 (closing). By Lemma 4, $\int \chi g^2 \le E_2/(c^* L_\ell)$, so by Cauchy–Schwarz $|\int g C_\chi| = |\int (\chi^{1/2} g)(\mathsf{Q} - \mathsf{P})| \le (E_2/(c^*L_\ell))^{1/2}(\delta_1 h + \rho_N)$ and $c^*|\int \chi g w| \le (E_2/(c^*L_\ell))^{1/2} c^* \tau_w$. Insert Step 4 into Proposition 3. The coefficient of E_2 is $\delta' = \delta_1 (t_- \rho_1)^{1/2}$, so with $e_{\mathrm{out}} = |\int_{|y|>A} g\, \varphi * (\chi \mathcal{M})|$,

$$E_2 \le \delta' E_2 + b E_2^{1/2} + e_{\mathrm{out}},$$
$$b = \frac{\delta_1 t_-^{1/2}(c^*\tau_w + \|F\|_{\mathrm{out}}) + \rho_N + c^*\tau_w}{(c^* L_\ell)^{1/2}}.$$

If $\delta' < 1$ then $E_2 \le \big[(b + (b^2 + 4(1-\delta')e_{\mathrm{out}})^{1/2})/(2(1-\delta'))\big]^2$, which is nondecreasing in b, e_{out} and δ'.

Step 6 (the edge terms are bounded). Let $\varepsilon = 2\Lambda_0/A \le 1$ and $d = A - |x|$; then $\chi \le e^{\varepsilon d} \le e^d$ on $[-A,A]$. First, $w \le 2\bar\Phi(d)$ gives $\tau_w^2 \le 8\int_0^\infty e^d \bar\Phi(d)^2 dd =: \bar\tau^2$. Second, for $x = A + t$ we have $0 \le \varphi * (c^* \mathbf{1}_{[-A,A]})(x) \le c^* \bar\Phi(t)$ and $0 \le p(x) \le e\, c^* e^{-t^2/2}$, so $\|F\|_{\mathrm{out}} \le c^* \bar F$ with $\bar F^2 = 2\int_0^\infty (\bar\Phi(t) + e\, e^{-t^2/2})^2 dt$. Third, at $y = A + t$ the continuous part of $\varphi * (\chi\mathcal{M})$ is at most $c^* \int_0^\infty e^s \varphi(t+s) ds$, and, covering by the intervals I_k and using Lemma 3(b), the atomic part is at most $P_{\mathrm{loc}} c^* \sum_{k\ge 0} e^{k+1}\varphi(t+k)$. With $|g| \le \lambda_1 + t^2/2 + \mu_B t$ this gives $e_{\mathrm{out}} \le c^* \bar e$ with a finite constant $\bar e$: indeed $\int_0^\infty e^s \varphi(t+s)\, ds \le e^{1/2 - t}$ and $\sum_{k\ge 0} e^{k+1}\varphi(t+k) \le (2\pi)^{-1/2} e^{1 - t^2/2} \sum_{k\ge 0} e^{k - k^2/2} < \infty$, so the product with the quadratic bound on $|g|$ is integrable over $t \ge 0$. The constants $\bar\tau, \bar F, \bar e$ do not depend on A or $\Lambda_0 \le A/2$.

Step 7 (asymptotics). Let $B^2 = \Lambda_0 + 4\log\Lambda_0$. As $A \to \infty$: $\gamma = \frac{c_0}{2} A^{-4/3} \to 0$, $\Delta = 4\gamma(1 + o(1))$, $\Delta B^2 = O(A^{-2/3})$, $\alpha_\pm(A)^2 = (1 \pm 2\gamma)^{-1} \exp(4\gamma^2 A^2/(1 \pm 2\gamma))$ with $4\gamma^2 A^2 = \Lambda_0^2/A^2 \to 0$, so $A_2 \to 1$; and $A_1 = O(\gamma)$, because the prefactors $(1 \mp 2\gamma)^{-1/2}$ differ by $2\gamma(1 + O(\gamma))$ and the exponents $2\gamma^2 A^2/(1 \mp 2\gamma)$ differ by $8\gamma^3 A^2/(1 - 4\gamma^2) = O(\gamma)\, O(\gamma^2 A^2)$. Therefore $\delta_1 = A\Delta B(1 + o(1)) = 2\Lambda_0 B/A\,(1 + o(1)) \to 2c_0^{3/2}$, because $\Lambda_0 B/A = c_0^{3/2}(1 + o(1))$; and $\delta' \to \delta_\infty := 2c_0^{3/2}\rho_1^{1/2}$, which is less than 1 exactly when $c_0 < c_{0,\max}$. For ρ_N: for $B \ge 1$ and $t \in [\frac12, 2]$ we have $R_k(t,B) \le c_k B^{2k} e^{-tB^2}$ with $c_k = \pi^{-1}\int_0^\infty (1+u)^{2k} e^{-u}\, du$, since $\xi = B + u$ gives $\xi^{2k} \le B^{2k}(1+u)^{2k}$ and $t(\xi^2 - B^2) \ge u$, and $t_+ B^2 \ge B^2 - 2\gamma B^2$ with $\gamma B^2 \to 0$, so $N e^{-t_+ B^2/2} \le 4e^{(\Lambda_0 - B^2)/2} = 4\Lambda_0^{-2}$ for large A. Hence each of the three terms of ρ_N is $O(\Lambda_0^{-1/2}/A)$ (the third, of size $\Delta A B \Lambda_0^{-2}$, is the largest) and, by Lemma 2, $\rho_N/c^* \le \rho_N e^{1/2}(2A + \sqrt{2\pi})$ is bounded. Thus for $A \ge A_0(c_0)$ we have $\delta' \le \bar\delta := \frac12(1 + \delta_\infty) < 1$, and $b/c^{*1/2}$ and e_{out}/c^* are bounded by constants depending only on c_0, which gives $E_2 \le \bar E c^*$. □

Corollary 1. *Under the hypotheses of Proposition 4, for* $A \ge A_0$ *and* $A - |x_0| \ge 1$ *we have* $|p(x_0)/c^* - 1| \le C_1 \exp(-\Lambda(x_0)/3)$ *with* $\Lambda(x_0) = \Lambda_0(1 - x_0^2/A^2)$ *and* $C_1 = [4L_{\mathrm{e}}(e-1)e^{0.1}\bar E]^{1/3}$.

Proof. Let $\eta = |p(x_0)/c^* - 1| \le e - 1$ and $\zeta = \eta/(2L_{\mathrm{e}}) \le 0.0919$. By Lemma 3(e), $|p - c^*| \ge \eta c^*/2$ on $I = [x_0 - \zeta, x_0 + \zeta] \subset [-A, A]$. On I we have $\chi \ge e^{-0.1} e^{\Lambda(x_0)}$ because $\Lambda_0(x^2 - x_0^2)/A^2 \le (2\Lambda_0/A)\zeta + \Lambda_0\zeta^2/A^2 \le \zeta + \zeta^2/2 < 0.1$ there, by $2\Lambda_0/A \le 1$ and $A \ge 1$. Since $(r-1)^2 \le (e-1)(r-1)\log r$, the integrand of E_2 is at least $(p - c^*)^2/((e-1)c^*)$, so $\bar E c^* \ge E_2 \ge e^{-0.1} e^{\Lambda(x_0)} \eta^3 c^*/(4L_{\mathrm{e}}(e-1))$. Solve for η. □

C. A local lemma of Toeplitz type

Lemma 5. *Let* μ *be a probability measure on* $\mathbb{R}$ *with density* $m = \varphi * \mu$*, let* $f_0 > 0$ *and let* $M \ge \sup m/f_0$*. Let* $W = [x_0 - L, x_0 + L]$ *be such that* μ *restricted to* W *is a sum of* $k < \infty$ *point masses (the atoms of* μ *in* W*), and let* $\eta > 0$ *satisfy* $\int_W (m - f_0)^2 \le \eta^2 f_0^2\, 2L$*. Fix* $\sigma > \sqrt{2}$ *and* $\bar\rho > 1$*; put* $s = \sqrt{\sigma^2 - 1}$*,* $\kappa_1 = \sigma^2/(\sigma^2 - 1)$*,* $\kappa_2 = \sigma^2/(\sigma^2 - 2)$*, and let* $\tau \in (0, L)$ *with* $l = L - \tau \ge 1$*, and put* $\omega_0 = \pi/l$*. Let* $B_0 = \frac12$*,* $B_n = 1 + (1 + \log 2n)/\pi$ *for* $n \ge 1$*, let* $I_l = 4 + 4\log l + 2(\pi/(\sigma^2/2 - 1))^{1/2}$*,*

$$f(n) = \frac{\bar\rho - 1}{\bar\rho}\,\frac{l}{\pi}\Big(\frac{2}{\kappa_2}\Big)^{1/2} \times \Big(\max\Big\{0, \log\frac{1}{2\eta(\bar\rho L B_n I_l/(\pi l))^{1/2}}\Big\}\Big)^{1/2},$$
$$n(j) = \Big\lceil \frac{\bar\rho j}{\bar\rho - 1} \Big\rceil - 1 \ (j \ge 1), \qquad n(0) = 0.$$

Assume that for an integer $n^* \ge 0$*:*

(a) $\bar\rho\big[\sigma/(\sqrt{2\pi}\, l) + \bar\Phi(\tau/\sigma)/(2l f_0)\big] \le \frac14$;
(b) $(1 + M)((2n^* + 1)\bar\rho)^{1/2}(s/l)\varphi(\tau/s)\, e^{\kappa_1 n^{*2}\omega_0^2/2} \le \frac14$.

If $k^* > 0$ *satisfies* $k^* \le f(n^*)$ *and* $n(\lceil k^* \rceil - 1) \le n^*$*, then* $k \ge k^*$.

Proof. Translate so that $x_0 = 0$. Let $n \ge k$ and let $V \subset \mathbb{C}^{n+1}$ be the space of coefficient vectors c whose polynomial $p_c(\theta) = \sum_{a=0}^n c_a e^{ia\omega_0\theta}$ vanishes at the k atoms in W. Then $\dim V \ge n + 1 - k \ge 1$ and $\rho_V := (n+1)/\dim V$. Let Π be the orthogonal projector onto V, and let $Q_V(\theta) = \sum_r |p_{v^{(r)}}(\theta)|^2$ for an orthonormal basis $v^{(r)}$ of V, so $Q_V = \sum_{|j|\le n} q_j e^{ij\omega_0\theta}$ with $q_j = \sum_{a - a' = j} \Pi_{aa'}$. Then $Q_V \ge 0$ vanishes at the atoms in W; $Q_V \le n+1$, because $Q_V(\theta) = \varepsilon(\theta)^* \Pi\, \varepsilon(\theta)$ with $\varepsilon(\theta) = (e^{-ia\omega_0\theta})_{a=0}^n$, $|\varepsilon|^2 = n + 1$ and Π a projector; $\int_{-l}^{l} Q_V = 2l \dim V$, because $\omega_0 l = \pi$ gives $\int_{-l}^{l} e^{ij\omega_0\theta} d\theta = 0$ for $j \ne 0$, so that only $q_0 = \operatorname{tr}\Pi = \dim V$ survives; and $\sum_j |q_j|^2 \le (n+1)\|\Pi\|_F^2 = (n+1)\dim V$, by Cauchy–Schwarz on each diagonal of Π, which has at most $n + 1$ entries.

Test function. Let $\mathrm{win} = \mathbf{1}_{[-l,l]} * \mathcal{N}(0, \sigma^2)$, so $\widehat{\mathrm{win}}(\xi) = \frac{2\sin l\xi}{\xi} e^{-\sigma^2\xi^2/2}$, and $F_t = Q_V\, \mathrm{win} \ge 0$. For real ν let $G_\nu(y) = e^{\nu^2/2} e^{i\nu y}[\Phi(\frac{y + l - i\nu}{s}) - \Phi(\frac{y - l - i\nu}{s})]$ and $G = \sum_j q_j G_{j\omega_0}$. Shifting the contour in $y \mapsto \int_{-l}^{l} \varphi_s(y - x - i\nu) dx$ gives $\hat G_\nu(\xi) = e^{\xi^2/2}\widehat{\mathrm{win}}(\xi - \nu)$ (the shift $y \mapsto y + i\nu$ gives the exponent $\nu^2/2 + \nu(\xi - \nu) - s^2(\xi - \nu)^2/2$, which equals $\xi^2/2 - \sigma^2(\xi - \nu)^2/2$ because $\sigma^2 = s^2 + 1$), hence $\varphi * G_\nu = e^{i\nu\cdot}\mathrm{win}$, $\varphi * G = F_t$ and $\int G = \int F_t$. Since $|\varphi_s(a - i\nu)| =$

$\varphi_s(a)e^{\nu^2/(2s^2)}$, we have $|G_\nu| \le e^{\kappa_1\nu^2/2}(\mathbf{1}_{[-l,l]} * \varphi_s)$. As $\int G\,m\,dy = \int(\varphi * G)\,d\mu$,

$$f_0\int F_t - \int F_t\,d\mu = \int G\,(f_0 - m). \tag{8}$$

Bounds. (i) $\int F_t \ge \int_{-l}^{l} Q_V - (n+1)\int_{-l}^{l}(1-\mathrm{win}) \ge 2l\dim V - (n+1)\,2\sigma/\sqrt{2\pi}$. (ii) $\int F_t\,d\mu \le (n+1)\bar\Phi(\tau/\sigma)$, since $Q_V = 0$ at the atoms in W and $\mathrm{win}(\theta) \le \bar\Phi(\tau/\sigma)$ for $|\theta| > L$. (iii) On W: $\int_W |G||f_0 - m| \le \|G\|\,\eta f_0(2L)^{1/2}$. (iv) Off W we have $|f_0 - m| \le (1+M)f_0$ and $|G| \le \big(\sum_j |q_j|\big)e^{\kappa_1 n^2\omega_0^2/2}(\mathbf{1}_{[-l,l]} * \varphi_s)$, while $\sum_j |q_j| \le ((2n+1)(n+1)\dim V)^{1/2}$ and $\int_{|y|>L}\mathbf{1}_{[-l,l]} * \varphi_s \le 2s\varphi(\tau/s)$; so the contribution is at most $(1+M)f_0((2n+1)(n+1)\dim V)^{1/2}e^{\kappa_1 n^2\omega_0^2/2}2s\varphi(\tau/s)$. (v) By Plancherel, $\|G\|^2 = \frac{1}{2\pi}\int e^{\xi^2}|\sum_j q_j\widehat{\mathrm{win}}(\xi - j\omega_0)|^2 d\xi$; Cauchy–Schwarz with the weights $|\widehat{\mathrm{win}}(\xi - j\omega_0)|$ and the two kernel estimates

$$\text{(K1)}\quad \sum_{|j|\le n}|\widehat{\mathrm{win}}(\xi - j\omega_0)| \le 4lB_n,$$

$$\text{(K2)}\quad \int e^{\xi^2}|\widehat{\mathrm{win}}(\xi-\nu)|\,d\xi \le e^{\kappa_2\nu^2}I_l.$$

They give $\|G\|^2 \le \frac{2lB_nI_l}{\pi}\sum_j |q_j|^2 e^{\kappa_2 j^2\omega_0^2} \le \frac{2lB_nI_l}{\pi}(n+1)\dim V\, e^{\kappa_2 n^2\omega_0^2}$. (K1): $|\widehat{\mathrm{win}}(\zeta)| \le \min(2l, 2/|\zeta|)$ and the points $\xi - j\omega_0$ are ω_0 apart; on each side of 0 the nearest point contributes at most $2l$ and the i-th point after it at most $2l/(i\pi)$, so the sum is at most $4l + \frac{4l}{\pi}\sum_{i=1}^{2n}\frac{1}{i} \le 4lB_n$ (and $\le 2l = 4lB_0$ for $n = 0$). (K2): with $u = \xi - \nu$ and $a = \sigma^2/2 - 1 > 0$ we have $e^{\xi^2 - \sigma^2u^2/2} = e^{\kappa_2\nu^2}e^{-a(u-\nu/a)^2}$; bound $|2\sin lu/u|$ by $2l$ for $|u| < 1/l$, by $2/|u|$ for $1/l \le |u| \le 1$ and by 2 for $|u| > 1$, and use $l \ge 1$.

Division. Divide (8) by $2lf_0\dim V$ and use $\rho_V \le \bar\rho$: the left side is at least $1 - \bar\rho[\sigma/(\sqrt{2\pi}\,l) + \bar\Phi(\tau/\sigma)/(2lf_0)] \ge \frac{3}{4}$ by (a); the contribution (iv) is at most $\beta(n) := (1+M)((2n+1)\bar\rho)^{1/2}(s/l)\varphi(\tau/s)e^{\kappa_1 n^2\omega_0^2/2}$; the contribution (iii) is at most $\eta(\bar\rho LB_nI_l/(\pi l))^{1/2}e^{\kappa_2 n^2\omega_0^2/2}$. Hence, if $\rho_V \le \bar\rho$ and $\beta(n) \le \frac{1}{4}$,

$$\kappa_2 n^2\omega_0^2/2 \;\ge\; \Xi_n := \log\frac{1}{2\eta(\bar\rho LB_nI_l/(\pi l))^{1/2}}. \tag{9}$$

Conclusion. Suppose $k < k^*$. If $k \ge 1$ take $n = n(k)$; then $n \ge k$ and $\rho_V \le (n+1)/(n+1-k) \le \bar\rho$ because $n+1 \ge \bar\rho k/(\bar\rho - 1)$, and $n(k) \le n(\lceil k^*\rceil - 1) \le n^*$, so $\beta(n(k)) \le \beta(n^*) \le \frac{1}{4}$ by (b). Then (9) holds at $n(k)$; as B_n is nondecreasing, $\Xi_{n(k)} \ge \Xi_{n^*}$, and $\Xi_{n^*} > 0$ because $0 < k^* \le f(n^*)$. Since $n(k) < \bar\rho k/(\bar\rho - 1)$, we get $\kappa_2(\bar\rho k/(\bar\rho - 1))^2\pi^2/(2l^2) > \Xi_{n^*}$, that is $k > f(n^*) \ge k^*$, a contradiction. If $k = 0$, take $n = 0$: $V = \mathbb{C}$, $\rho_V = 1$ and $\beta(0) \le \beta(n^*) \le \frac{1}{4}$. The division step with $\rho_V = 1$ in place of $\bar\rho$ in (iii) gives $0 \ge \log\frac{1}{2\eta(LB_0I_l/(\pi l))^{1/2}} \ge \Xi_0 \ge \Xi_{n^*} > 0$, a contradiction. □

D. Proof of Theorem 1

Fix $c_0 \in (0, c_{0,\max})$, a value $\sigma > \sqrt{2}$ and $\bar\rho > 1$, and let A be large. Let $\delta \in (0,1)$ and $L_w = A^{2/3}$. The windows $W_j = [x_j - L_w, x_j + L_w]$, $x_j = j(2L_w+1)$, are disjoint, and we keep those with $u_j := 1 - ((|x_j| + L_w)/A)^2 \ge \delta$; then $W_j \subset [-A\sqrt{1-\delta}, A\sqrt{1-\delta}] \subset [-A, A]$. By Proposition 4, $\chi \ge e^{\Lambda_0 u_j}$ on W_j and Lemma 4,

$$\begin{aligned}\int_{W_j}(p - c^*)^2 &\le e^{-\Lambda_0 u_j}(e-1)c^*E_2\\ &\le e^{-\Lambda_0 u_j}(e-1)\bar E\,c^{*2} = \eta_j^2 c^{*2}\,2L_w,\end{aligned}$$

with $\log(1/\eta_j) = \Lambda_0 u_j/2 + \frac{1}{2}\log(2L_w) - \frac{1}{2}\log((e-1)\bar E) =: \Theta_j$. Apply Lemma 5 to $\mu = P$, $f_0 = c^*$, $M = e$ (Lemma 3(a)) and $W = W_j$, with margin $\tau = s(2E_b + 2\log A)^{1/2}$, $E_b = \frac{\kappa_1}{2}\big((2\Theta_j/\kappa_2)^{1/2} + 2\pi\bar\rho/((\bar\rho-1)L_w)\big)^2$. Since $\Theta_j \ge \frac{c_0\delta}{2}A^{2/3} - O(1)$ and $\Theta_j \le \frac{c_0}{2}A^{2/3} + O(\log A)$ uniformly in j, we have $\tau = O(A^{1/3}) = o(L_w)$ and $l = L_w(1 - o(1))$.

Let $n_1 = n(\lceil f(0)\rceil)$ and $k_j^* = f(n_1)$. The logarithm in f is at most Θ_j (because $\bar\rho LB_0I_l/(\pi l) \ge 2\bar\rho/\pi > \frac{1}{4}$), so $f(0) \le \frac{\bar\rho - 1}{\bar\rho}\frac{l}{\pi}(2\Theta_j/\kappa_2)^{1/2}$ and, using $l \ge L_w/2$, we get $n_1\omega_0 \le (2\Theta_j/\kappa_2)^{1/2} + 2\pi\bar\rho/((\bar\rho-1)L_w)$, that is $\kappa_1 n_1^2\omega_0^2/2 \le E_b$. The logarithm in $f(n)$ equals $\Theta_j - \log 2 - \frac{1}{2}\log(\bar\rho LB_nI_l/(\pi l))$, and for $n \le n_1 = O(A)$ we have $B_n = O(\log A)$ and $I_l = O(\log A)$; so it is at least $\frac{c_0\delta}{2}A^{2/3} - O(\log\log A) > 0$ for large A. Hence $f(0) > 0$, so $\lceil f(0)\rceil \ge 1$, and $k_j^* > 0$. Also $k_j^* \le f(0)$, so $n(\lceil k_j^*\rceil - 1) \le n_1$. Condition (b) holds for large A: by the choice of τ we have $\tau^2/(2s^2) = E_b + \log A$, so $\varphi(\tau/s)e^{\kappa_1 n_1^2\omega_0^2/2} \le (2\pi)^{-1/2}A^{-1}$, while $((2n_1+1)\bar\rho)^{1/2}s/l = O(A^{1/2}A^{-2/3})$ because $n_1 = O(L_w\Theta_j^{1/2}) = O(A)$; so the left side of (b) is $O(A^{-7/6})$. Condition (a) holds for large A: the first term is $O(A^{-2/3})$, and, since $\tau^2/(2\sigma^2) \ge \Theta_j/\kappa_2$, we have $\bar\Phi(\tau/\sigma) \le \frac{1}{2}e^{-\Theta_j/\kappa_2}$ while $1/(2lc^*) \le e^{1/2}(2A + \sqrt{2\pi})/L_w = O(A^{1/3})$. Hence for large A every kept window is admissible, uniformly in j, and Lemma 5 gives at least k_j^* atoms in W_j. Now $B_{n_1} = O(\log A)$ and $I_l = O(\log A)$, so

$$k_j^* = \frac{\bar\rho - 1}{\bar\rho}\,\kappa_2^{-1/2}\,\frac{l}{\pi}\,(\Lambda_0 u_j)^{1/2}(1 - o(1)) \quad \text{uniformly in } j.$$

For $x \in W_j$ we have $1 - x^2/A^2 - u_j \le 4L_w/A$, so $\sqrt{u_j} \ge \sqrt{1 - x^2/A^2} - 2(L_w/A)^{1/2}$. Every x with $|x| \le A\sqrt{1-\delta} - 2L_w$ lies in a kept window or in one of the gaps of length 1 between the windows; there are $O(A^{1/3})$ windows, so the gaps cost $O(A^{1/3})$, and the term $2(L_w/A)^{1/2}$ costs at most $4A(L_w/A)^{1/2} = O(A^{5/6})$. Both are $o(A)$. Summing over disjoint windows, and using that $K(A)$ is an integer, so that W_j holds at least $\lceil k_j^*\rceil \ge k_j^*$ atoms,

$$\begin{aligned}K(A) &\ge \sum_j k_j^*\\ &\ge \frac{\bar\rho - 1}{\bar\rho}\kappa_2^{-1/2}\frac{\Lambda_0^{1/2}}{2\pi}\\ &\quad\times\int_{|x|\le A\sqrt{1-\delta}-2L_w}\Big(1 - \frac{x^2}{A^2}\Big)^{1/2}dx\,(1 - o(1)).\end{aligned}$$

Divide by $A^{4/3}$ and let $A \to \infty$; the integral is $A\int_{|y|\le\sqrt{1-\delta}}(1-y^2)^{1/2}dy\,(1+o(1))$. For fixed $(c_0, \sigma, \bar\rho, \delta)$ this is a lower bound for $\liminf K(A)/A^{4/3}$ that does not depend on A (the thresholds A_0 and $\bar E$ may depend on the parameters). Now let $\delta \to 0$ (the integral over y tends to $\pi/2$), then $\sigma, \bar\rho \to \infty$ and $c_0 \uparrow c_{0,\max}$: $\liminf_{A\to\infty} K(A)/A^{4/3} \ge \sqrt{c_{0,\max}}/4 = \frac{1}{4}(4\rho_1)^{-1/6}$. By the ball enclosure of Lemma 3(c) and $\rho_1 =$

$(e-1)\lambda_1/(1-e^{-\lambda_1})$, $\rho_1 \in 2.909027895 \pm 10^{-9}$ and $\sqrt{c_{0,\max}}/4 \in 0.16607542 \pm 10^{-8}$, which exceeds 0.1660. □

APPENDIX B
THE ONE-MINIMUM LEMMA

Let $P = \sum_{j=1}^{K} w_j \delta_{x_j}$ with $K \geq 3$ be the capacity-achieving input of the channel $Y = X + Z$ with $|X| \leq A$, let $Q = \varphi * P$ be its output density, and let $D(x) = \int \varphi(y-x) \log\big(\varphi(y-x)/Q(y)\big) dy$ be the optimality function with capacity C. We use two facts from the literature. (H1) By Smith [2], P is unique and finite, and $D \leq C$ on $[-A, A]$ with equality at the atoms. The channel and the constraint are invariant under $x \mapsto -x$, so the mirror image of P is also optimal, and by uniqueness P is symmetric. (H2) By Theorem 2 of Barletta and Dytso [17], $D < C$ on $[-A, A]$ away from the atoms, except possibly at $x = 0$.

Lemma 6. *Let s be the number of sign changes of D' on $(0, \infty)$. Then all of them lie in $(0, A)$ and $K-2 \leq s \leq K-1$. Here $D'(0+) > 0$ means $D' > 0$ on $(0, \varepsilon)$ for some $\varepsilon > 0$, and similarly for $D'(0+) < 0$; one of the two holds because D' is analytic, odd and not identically zero. If K is odd, or if K is even and $D'(0+) > 0$, then $s = K-2$. If K is even and $D'(0+) < 0$ (the origin is a local maximum of D), then $s = K-1$. Equivalently, $D - C$ has exactly one local minimum between consecutive atoms, except in the central gap of an even K, where it has one minimum at 0 or two minima at $\pm z_0$ with a maximum at 0.*

Proof. Preliminaries. The output density $Q > 0$ is analytic and $Q'/Q = -y + \mathbb{E}[X \mid Y = y]$, with the posterior mean strictly inside $(-A, A)$. Differentiating under the integral, $D'(x) = -(\varphi * (Q'/Q))(x) = x - \int \varphi(y-x)\, \mathbb{E}[X \mid Y = y]\, dy$. So $D'(x) > x - A$, and $D' > 0$ for $x \geq A$. The posterior mean is also below the largest atom a, so $D'(x) > x - a \geq 0$ for $x \geq a$. If $a < A$, then D would increase strictly on $[a, A]$ from $D(a) = C$ and exceed C, against $D \leq C$. So A is an atom, and by the symmetry so is $-A$. The function D is analytic and D' is odd, so D' changes sign at 0.

Two tools. (E) A real exponential polynomial $f(x) = \sum_{j=1}^{n} P_j(x) e^{\lambda_j x}$ with distinct real λ_j and nonzero polynomials P_j of degree d_j has at most $N-1$ real zeros counted with multiplicity, $N = \sum_j (d_j + 1)$. Induct on N. For $n = 1$ the zeros are those of P_1. For $n \geq 2$ put $g = e^{-\lambda_n x} f$ and differentiate $d_n + 1$ times: the term P_n disappears, the others keep their degrees, and by Rolle's theorem g has at most $d_n + 1$ more zeros than $g^{(d_n+1)}$. (V) The Gaussian kernel is strictly totally positive, so it does not increase the number of sign changes of a function of polynomial growth [29], [39].

Upper bound. The derivative $Q'(x) = \varphi(x) \sum_j c_j (x_j - x) e^{x_j x}$ with $c_j = w_j e^{-x_j^2/2} > 0$ is, up to the positive factor φ, an exponential polynomial with K exponents and degree-one coefficients, so $N = 2K$. By (E), Q' has at most $2K-1$ real zeros, and so does Q'/Q. By (V), $D' = -\varphi * (Q'/Q)$ has at most $2K-1$ sign changes. Since D' is odd with a sign change at 0, this is $2s + 1 \leq 2K - 1$, so $s \leq K - 1$.

Lower bound. Number the positive atoms $0 < u_1 < \cdots < u_m = A$. By (H1) and (H2), each atom u_j with $j < m$ is a strict local maximum of the analytic function D, so D' changes sign there from $+$ to $-$: $m - 1$ sign changes. In each gap between consecutive points of $\{0 \text{ (if an atom)}, u_1, \ldots, u_m\}$, $D - C < 0$ inside and 0 at the end points, with $D'(A) > 0$ at the last. So D has a minimum in the gap, where D' changes sign from $-$ to $+$. For odd $K = 2m + 1$ there are m gaps, so $s \geq 2m - 1 = K - 2$. For even $K = 2m$ there are $m - 1$ gaps between positive atoms, so $s \geq 2m - 2 = K - 2$.

Parity. The derivative satisfies $D' > 0$ on (A, ∞), so the sign of $D'(0+)$ equals $(-1)^s$. For odd K we have $D'(0+) < 0$ (the atom at 0 is a local maximum), so s is odd and $s = K-2$. For even K with $D'(0+) > 0$, s is even, so $s = K - 2$. For even K with $D'(0+) < 0$, s is odd and the bounds leave only $s = K - 1$. The extra sign change lies in $(0, u_1)$. Indeed $D' < 0$ just to the right of 0, and $D' > 0$ just to the left of u_1, which is a strict local maximum of D. With the $K - 2$ changes counted above, all $K - 1$ are accounted for, so D has one minimum at $\pm z_0$ in the central gap and a maximum at 0. For even K with $D'(0+) > 0$ the $K - 2$ counted changes are all of them, so D' does not change sign in $(0, u_1)$ and D has its minimum at 0. □

For even K the case $D'(0+) < 0$ means that D has a local maximum at the origin; Barletta and Dytso illustrate such a contact, with $D''(0) < 0$, at the birth of an atom at the origin [17]. The stored optimal states end before this case: the center signal $D(0) - C$ is down to -2×10^{-18} at $K = 82$ and $D'(0.01) > 0$. At 25 other states certified in ball arithmetic, the middle state by A of each family $K = 40$ and 60 to 83, the number of sign changes of D' on $(0, A)$ is $K - 2$. The same holds at the last eight optimal states of each of the even-K families $K = 60$, 62, 64, 66, 80 and 82.

APPENDIX C
PROOF OF THEOREM 2

Here φ is the standard normal density. All functions are even in y, and so is $1/Q$. Write $\varphi^s(y; u) = \varphi(y-u) + \varphi(y+u)$. The value functions are $\varphi_j = \varphi^s(\cdot; u_j)$, $\varphi_A = \varphi^s(\cdot; A)$ and, for odd K, also $\varphi_0 = \varphi$. Then $Q = \sum_j w_j \varphi_j + w_e \varphi_A \,(+ w_0 \varphi_0)$ with positive masses. The derivative functions are $\psi_p = w_p\, \partial_u \varphi^s(\cdot; u_p)$ and $\psi_A = w_e\, \partial_A \varphi^s(\cdot; A)$. The mass directions are $\chi_k = \varphi_k - \varphi_A$ and, for odd K, also $\chi_0 = \varphi_0 - \varphi_A/2$. Let U be their span and N the Gram matrix of $\psi_p - P_U \psi_p$, $p \in \{1, \ldots, m, A\}$, with P_U the orthogonal projection onto U, for the inner product $\langle f, g \rangle = \int fg/Q$. The value functions are bounded by Q/w with the mass w of their atom, and $|\psi_p| \leq (|y| + u_p)\, w_p \varphi_p$. So every product of two of these functions, divided by Q, is at most a polynomial times a Gaussian, and all the integrals below exist.

Step 1: the kernel. In the variables $s = y^2$ and $t = u^2$, $\varphi^s(y; u) = \frac{2}{\sqrt{2\pi}} e^{-s/2} e^{-t/2} F(st)$ with the entire function $F(z) = \sum_k z^k/(2k)!$. Take rows $0 < y_1 < \cdots < y_n$ and columns forming a Hermite pattern in the node variable: nodes $0 \leq u_1 < \cdots < u_r$, the value column at every node and the derivative column at some nodes with $u_i > 0$. Expand every entry in the series. The sums converge absolutely, so the Cauchy–Binet formula applies over the exponents. The

determinant is a sum of terms with positive coefficients, each a generalized Vandermonde determinant in s times a confluent determinant of monomials in t. Both are positive, because monomials form an extended Chebyshev system on $(0,\infty)$. At a node $t=0$ (the center atom of an odd K) the column has the single entry 1 for the exponent 0, so only exponent sets that contain 0 contribute. Each of them leaves a determinant of the same kind on the other nodes, and at least one such set exists. The derivative of an entry in u is $2u\,e^{-t/2}\big(s\,F'(st)-\frac{1}{2}F(st)\big)$ times the other positive factors. The factors $2u$, $e^{-s/2}$ and $e^{-t/2}$ are positive scalings. After dividing the derivative column by $2u$, the term $-\frac{1}{2}F$ is minus one half of the value column of the same node, which is in every Hermite pattern, and adding half of the value column to the derivative column does not change the determinant. At the node 0 the value function $\varphi_0=\varphi=\frac{1}{2}\varphi^s(\cdot;0)$ is a positive multiple of the kernel. So all minors of Hermite patterns of this form are positive. This is the extended total positivity of Karlin [29], in the form used here.

Step 2: Gram minors. Let I and J be Hermite patterns of the same size k. By the basic composition formula, the minor $\det[\langle f_a,f_b\rangle]_{a\in I,b\in J}$, where f_a is the value or derivative function of the index a, equals 2^k times the integral over $0<y_1<\cdots<y_k$ of $\det[f_a(y_t)]\det[f_b(y_t)]\prod_t 1/Q(y_t)$. Both determinants are positive by Step 1, and the integrand is continuous, so the minor is positive.

Step 3: the mass directions. Every reflected atom integrates to 2 and φ_0 integrates to 1, so each χ_k integrates to 0. By Step 2 with $I=J$ the set of all value functions, their Gram determinant is positive, so they are linearly independent. Hence V, their span, has one more dimension than U. The density Q lies in V, and $\langle Q,f\rangle=\int f$. So Q is orthogonal to U and not in U, because $\langle Q,Q\rangle=\int Q>0$, and V is the orthogonal sum of U and the line of Q. Every derivative function integrates to 0, hence it is orthogonal to Q, and $P_U\psi_p=P_V\psi_p$, with P_V the orthogonal projection onto V. Therefore N is the Schur complement of the Gram matrix over the block of all value functions.

Step 4: the minors of N. By Schur's determinant formula, $\det N[\alpha,\beta]=\det G[(\alpha,\gamma),(\beta,\gamma)]/\det G[\gamma,\gamma]$. Here G is the Gram matrix of the value and derivative functions, γ is the block of values, and α comes first in the order of the rows and columns. The denominator is positive by Step 2, with $I=J=\gamma$. Reorder rows and columns to the natural order of the nodes, value before derivative at each node. Each derivative function passes $c(p)$ value functions, which gives the sign $(-1)^{c(p)}$ per element of α and of β. Because γ holds the value of every node, including A and, for odd K, the node 0, both reordered index sets are Hermite patterns. So the numerator is positive by Step 2. Hence $\det(E'NE')[\alpha,\beta]>0$ for all α and β of equal size. □

Step 5: the link to the reduced Hessian. Steps 5 and 6 are not used in the proof of Theorem 2; they give the two identities of Section V-B. This step is a calculation at a stationary configuration, where $D'(u_j)=0$ for all j. Let $D(x)$ be the relative entropy of the unit normal centered at x from Q. For a parameter θ of Q we have $\partial_\theta D(x)=-\int\varphi(y-x)\,\partial_\theta Q/Q\,dy$. Since Q and $\partial_\theta Q$ are even in y, this equals $-\frac{1}{2}\langle\varphi^s(\cdot;x),\partial_\theta Q\rangle$. Also $\partial_x\varphi^s(\cdot;u_p)=\psi_p/w_p$, $\partial_{u_q}Q=\psi_q$ and $\partial_{w_k}Q=\chi_k$, where the mass constraint is built into the directions χ_k. The unknowns are the positions u_q and the masses w_k of the positive atoms (and w_0 for odd K), and the wall mass w_e follows from the total mass, so that $\partial Q/\partial w_k=\chi_k$. So the Jacobian of the rows $D'(u_p)$ and $D(u_j)-D(A)$ with respect to the columns u_q and w_k is $-\frac{1}{2}\operatorname{diag}(1/w_p,1)$ times the Gram matrix of the functions ψ_q and χ_k. For the centre atom of an odd K the row $D(0)-D(A)$ has the factor -1 in place of $-\frac{1}{2}$, because $\varphi_0=\varphi$ is not symmetrised. A row scale does not change the Schur complement below. The only other term is $D''(u_p)$ on the diagonal of the position block. Off stationarity the row $D(u_j)-D(A)$ also gets the term $D'(u_j)$, so this identity needs stationarity. The row scaling is block diagonal. By Step 3 the Schur complement over the mass block is $-\frac{1}{2}\operatorname{diag}(1/w_p)$ times N, so $S_G=\operatorname{diag}(c)\,N$ with $c_p=-1/(2w_p)$. The same computation for the wall column gives the Gram part $b_G=c\,N_{\cdot A}$ of the forcing. The rows $D(u_j)-D(A)$ add the term $-D'(A)$ of the moving evaluation point A, which is not of Gram type.

Step 6: the diagonal term. Since $\int\varphi\log\varphi$ does not depend on x, we have $D(x)=-h_1-\int\varphi(y-x)\log Q(y)\,dy$. One integration by parts gives $D'(x)=-\int\varphi(y-x)\,(Q'/Q)(y)\,dy$, and differentiating under the integral, with $\partial_x\varphi(y-x)=(y-x)\varphi(y-x)$, gives $D''(x)=-\int(y-x)\varphi(y-x)\,(Q'/Q)(y)\,dy$. The boundary terms vanish and the integrals converge, because $|Q'/Q|=|-y+\mathbb{E}[X\mid Y=y]|\le|y|+A$ and $|\log Q|$ grows at most quadratically. The score Q'/Q is odd in y, so only the odd part of $(y-x)\varphi(y-x)$ contributes, and that odd part is $\frac{1}{2}\psi^o(y;x)$ with $\psi^o(y;u)=(y-u)\varphi(y-u)+(y+u)\varphi(y+u)=\partial_u[\varphi(y-u)-\varphi(y+u)]$. With x_k and ν_k all atoms and masses, mirror images included, and grouping the atoms in mirror pairs, $Q'(y)=-\sum_k\nu_k(y-x_k)\varphi(y-x_k)=-\sum_q\tilde{w}_q\psi^o_q(y)$, where a center atom gives $-w_0y\varphi(y)=-\frac{w_0}{2}\psi^o_0(y)$. Hence $D''(u_p)=-\frac{1}{2}\langle\psi^o_p,Q'\rangle=\frac{1}{2}\sum_q\tilde{w}_q\langle\psi^o_p,\psi^o_q\rangle$. No step uses the Kuhn–Tucker conditions, so the identity holds at every configuration with positive masses.

Appendix D
Proof of Proposition 1

Fix one of the states, with amplitude A, positions $u_1<\cdots<u_m$, masses w_j, the wall mass w_e and, for odd K, the center mass w_0. Write x_k, ν_k for all atoms and masses, mirror images included, and $Q(z)=\sum_k\nu_k\varphi(z-x_k)$. Let $a=1/2$, $h=0.01$ and $Y=A+40$. Every quantity below was evaluated in ball arithmetic (Arb, through python-flint): at 192 bits, at 600 bits for the polish and the Krawczyk step, and at 1200 bits for the minors. All balls are taken over the whole box $\mathbf{X}$ of radius $r=10^{-110}$ around the polished state. For the stationary state with $K=80$ the box has radius $r=10^{-150}$ and the grid step is $h=0.005$; over the box of radius 10^{-110} some of its smallest minors cannot be given a sign. Steps 1 to 4 concern every state of Proposition 1, Step 5 its optimal states, and Step 6 those of them with odd K.

Step 1: a strip without zeros. For $z=y+ib$, $\varphi(z-x_k)/\varphi(z-x_1)=\rho_k(y)\,e^{ib(x_k-x_1)}$ with $\rho_k(y)=e^{(x_k-x_1)(y-(x_k+x_1)/2)}$. Choose a reference atom x_1 for a box

of y, and let S be the atoms other than x_1 with $a|x_k - x_1| \le \pi/2$. For $|b| \le a$ each term of S has a nonnegative real part, so

$$Q(z) = \nu_1 \varphi(z - x_1)\Big[1 + \sum_{k \in S} r_k e^{ib(x_k - x_1)} + R(z)\Big],$$
$$|R(z)| \le \tau(y) = \sum_{k \notin S \cup \{1\}} \frac{\nu_k}{\nu_1} \rho_k(y),$$

where $r_k = (\nu_k/\nu_1)\rho_k(y)$, and the bracket has real part at least $1 - \tau(y)$. On every box of width 0.01 of $[0, Y]$ the computed τ is below 1 (at most 0.187); by symmetry the same holds on $[-Y, 0]$. For $|y| \ge Y$ take the wall atom as reference: each ρ_k then decreases in $|y|$, and $\tau_A(Y) < 10^{-90}$ at each state. With this reference $r_S \le \sum_k \nu_k/w_e \le 1/w_e$, because $\rho_k \le 1$ for $|y| \ge Y$. Hence Q has no zero on $|\operatorname{Im} z| \le a$, $\log Q$ is analytic there, and $|\log Q(z)| \le |\log \nu_1| + \frac{1}{2}((|y - x_1| + a)^2 + a^2) + \log\sqrt{2\pi} + \max(|\log(1 - \tau)|, \log(1 + r_S + \tau)) + \pi/2$, with $r_S = \sum_{k \in S} r_k$.

Step 2: the error of the trapezoid sums. Each integral of the system and of its Jacobian has an integrand f analytic on the strip. By Theorem 5.1 of Trefethen and Weideman [34], $|\int f - h\sum_{n \in \mathbb{Z}} f(nh + c)| \le 2M/(e^{2\pi a/h} - 1)$, with M the largest integral of $|f|$ along a horizontal line of the strip. Step 1 bounds $|f|$ box by box on $|y| \le Y$. For $|y| \ge Y$ every integrand is at most

$$g(y) = e^{a^2/2}\varphi(|y| - A)\, 720\, (2|y|)^6 \times \Big(c_E + \log\sqrt{2\pi} + 3 + 3y^2 + \frac{2|y|}{w_e(1 - \tau_A(Y))}\Big),$$

where $c_E = |\log w_e| + \log\sqrt{2\pi} + a^2/2 + \max(|\log(1 - \tau_A(Y))|, \log(2 + 1/w_e)) + \pi/2$ bounds $|\log Q| - (|y| - A)^2/2$ there, and $g(y) \le g(Y)e^{-39(|y| - Y)}$. The integrands are $\mathrm{He}_k(z - x)\varphi(z - x)\log Q(z)$ and $\mathrm{He}_k(z - x)\varphi(z - x)\log\varphi(z - x)$ with $k \le 6$ and $\mathrm{He}_0 = 1$, and the Gram and forcing integrands $\mathrm{He}_k(z - u)\varphi(z - u)\,(z - x_i)\varphi(z - x_i)/Q(z)$. In g the factor $720\,(2|y|)^6$ bounds $|\mathrm{He}_k(z - x)|$ by the bound of Step 5, since $1 + |z - x| \le 2|y|$. The bracket bounds $|\log Q|$, $|\log\varphi|$ and $|z - x_i|\,|\varphi(z - x_i)/Q(z)|$, the last because $\rho_k \le 1$ there. The tails therefore add at most $2g(Y)/39$ to M, and the grid points with $|y| > Y$, which the computed sums omit, at most $2g(Y)(1/39 + h)$. With these terms the error of one sum is at most ε_F for the sums of the residual rows and ε_J for the Gram sums of the Jacobian, uniformly on $\mathbf{X}$ (at most 4.2×10^{-131} and 5.0×10^{-134}). A residual row $D(u_j) - D(A)$ is a difference of two sums, and a Jacobian entry or an entry of the forcing column is at most two Gram sums and one sum of the kind of D' or D''; we widen them by $2\varepsilon_F$ and by $2\varepsilon_J + \varepsilon_F$.

Step 3: one solution in the box. Let v_0 be the polished state, Y_0 the inverse of the midpoint of the computed Jacobian at v_0, and $K(\mathbf{X}) = v_0 - Y_0(F_h(v_0) + E_F) + (I - Y_0(J_h(\mathbf{X}) + E_J))(\mathbf{X} - v_0)$, with F_h, J_h the trapezoid sums and E_F, E_J the error terms added in Step 2. At every state $K(\mathbf{X})$ lies in the interior of $\mathbf{X}$ (largest $|K(\mathbf{X}) - v_0|$ is below 3.1×10^{-116}), so $\mathbf{X}$ contains exactly one solution of the Kuhn–Tucker system with exact integrals [35]–[37]; all its masses are positive.

Step 4: the minors. The matrix S and the vector b of Section V-B are formed from $J_h(\mathbf{X}) + E_J$, and the $m(m+1)$ initial minors of $X = [\,T \mid v\,]$ are certainly positive as balls over $\mathbf{X}$. They are therefore positive at the solution of Step 3, X is strictly totally positive [30, Thm. 4.1], and Lemma 1 gives $du_i/dA > 0$ at that state, for the branch of stationary states through it. The branch exists by the implicit function theorem, because the inclusion of $K(\mathbf{X})$ in the interior of $\mathbf{X}$ also shows that the Jacobian of the system is nonsingular on $\mathbf{X}$ [37].

Step 5: optimality. Since $\int \varphi\log\varphi$ does not depend on x, we have $D(x) - D(A) = G(A) - G(x)$ with $G(x) = \int \varphi(y - x)\log Q(y)\,dy$, and $G^{(k)}(x) = \int \mathrm{He}_k(y - x)\varphi(y - x)\log Q(y)\,dy$. These are computed as sums over $|y - x| \le 14$, with the error of Step 2 and the bound $|\mathrm{He}_k(s)| \le k!(1 + |s|)^k$ for $k \le 6$. It holds for complex s: the coefficient $k!/(j!\,(k - 2j)!\,2^j)$ of s^{k-2j} in He_k is at most the coefficient $k!\binom{k}{2j}$ of $|s|^{k-2j}$ in $k!(1 + |s|)^k$, because their ratio is $(2j - 1)!!/k! \le 1$. The grid points outside the window are bounded by Gaussian tails, through the largest $|\log Q|$ on the grid for $|y| \le Y$ and through g for $|y| > Y$ (at most 1.4×10^{-31}). At the solution of Step 3, $D(u_j) = D(A)$ and $D'(u_j) = 0$. Around each positive atom, and around 0 for odd K, there is a half-width $\zeta > 0$ for which the bound $-G''(u) + |G'''(u)|\zeta + \sup|G''''|\zeta^2/2 < 0$ on $[u - \zeta, u + \zeta]$ gives $D'' < 0$ there, hence $D \le D(A)$; near the wall, $G'(A) + |G''(A)|\zeta + \sup|G'''|\zeta^2/2 < 0$ gives $D' > 0$ on $[A - \zeta, A]$. The rest of $[0, A]$ is covered by boxes on which a Taylor polynomial of order 5 with the remainder of order 6 gives $D - D(A) < 0$ (at least 1.1×10^{-29} below zero). By symmetry $D \le D(A)$ on $[-A, A]$ with equality only at the atoms, so the solution satisfies the Kuhn–Tucker conditions of [2] and is the capacity-achieving input, which is unique. The inequality is strict off the atoms and $D''(u_j) < 0$ and $D'(A) > 0$. These inequalities persist for nearby amplitudes. Let $A' \mapsto v(A')$ be the branch of Step 4, and $D_{A'}$ the function D of its state. The zones and the gap boxes are fixed compact sets, and $D_{A'}$ and its first two derivatives in x are jointly continuous in (x, A'). Since $D' > 0$ on the compact interval $[A - \zeta, A]$, there is $\delta > 0$ with $D' > 0$ on $[A - \zeta - \delta, A + \delta]$. So for $|A' - A| < \delta$ small enough $D''_{A'} < 0$ on every atom zone, which then contains the atom $u_j(A')$ with $D'_{A'}(u_j(A')) = 0$; $D'_{A'} > 0$ on $[A - \zeta - \delta, A + \delta]$, which contains $[A' - \zeta, A']$; and $D_{A'} - D_{A'}(A') < 0$ on every gap box. Hence $D_{A'} \le D_{A'}(A')$ on $[-A', A']$, the state $v(A')$ is the capacity-achieving input at A', and its positive atoms move outward by Step 4.

Step 6: the center mass. For odd K the branch of Step 4 has $dv/dA = -J^{-1}b$, with J the Jacobian of the Kuhn–Tucker system and b its derivative in A. Both are enclosed over $\mathbf{X}$ by $J_h(\mathbf{X}) + E_J$ and by the trapezoid sums of b widened as in Step 2, and J is nonsingular there by Step 3. Solving the linear system in ball arithmetic at 600 bits gives an enclosure of dv/dA at the exact state. At every state with odd K the enclosure of dw_0/dA is positive, and every enclosed du_i/dA is positive. At these states this proves the outward motion a second time, without the minors of Step 4. □

APPENDIX E
CONSTANTS OF THE CERTIFICATE

The certificate of Section VII evaluates the relative entropy $D_\pi(q)$ by a trapezoid rule on a grid, and a rigorous bound on the error of that rule is needed for complex q as well. The next lemma gives this bound with a complete proof. It uses two assumptions that the atoms satisfy in the application: they lie within the radius r_p, and their weights sum to at least $1/4$ (in the application the sum is 1 up to the radius of the enclosure box).

Lemma 7. *Let $x_1,\dots,x_n \in \mathbb{R}^2$ with $|x_k| \le r_p$ and weights $w_k > 0$ with $s = \sum_k w_k \ge 1/4$. Put $S(y) = \sum_k w_k \exp(\langle y, x_k\rangle - |x_k|^2/2)$ for $y \in \mathbb{C}^2$, and put $\varphi(u) = (2\pi)^{-1/2} e^{-u^2/2}$ and $\varphi_2(y) = \varphi(y_1)\varphi(y_2)$. Let $0 < a\, r_p < \pi/2$ and $0 < h \le 1$, let $\sigma \ge 0$ and $Q_1, Q_2 \ge 0$, and let $q \in \mathbb{C}^2$ satisfy $|\operatorname{Re} q_j| \le Q_j$ and $|\operatorname{Im} q_j| \le \sigma$. Define*

$$F(y) = \varphi_2(y-q)\log S(y), \qquad I = \int_{\mathbb{R}^2} F,$$
$$T_h F = h^2 \sum_{(m,n)\in\mathbb{Z}^2} F(mh, nh),$$
$$T_h^N F = h^2 \sum_{|m|,|n|\le N} F(mh, nh),$$

with $Y = Nh$, $\eta = 3e^{-2\pi^2/h^2}$, $E = \exp\big((a+\sigma)^2/2 + \sigma^2/2\big)$, $A_0 = r_p^2/2 + |\log\cos(a r_p)| + a r_p + 2|s-1|$ and $B_0 = r_p$. Then:

(i) *For real y, every $j \in \{1,2\}$ and real $|b| < a$ we have $\operatorname{Re} S(y + ibe_j) > 0$. So $\log S$ is holomorphic in y_j on the strip $|\operatorname{Im} y_j| < a$, the other coordinate being real, and $|\log S(y+ibe_j)| \le A_0 + B_0(|y_1| + |y_2|)$.*

(ii) *$|I - T_h F| \le 4(1+\eta)\, E\,[A_0 + B_0(Q_1 + Q_2 + 1.6 + 0.484\,h)] \,/\, (e^{2\pi a/h} - 1)$.*

(iii) *If $Y - Q_j \ge 2$ for $j = 1, 2$, then $|T_h F - T_h^N F| \le e^{\sigma^2} \sum_{(j,l)=(1,2),(2,1)} \big[(1+\eta)\tau_1(Q_j) + B_0\tau_0(Q_j)\big((1+\eta)Q_l + 0.7979 + 0.484h\big)\big]$, where $\tau_0(c) = 2\psi(Y-c)/(Y-c)$, $\tau_1(c) = 2\psi(Y-c)\,[(A_0 + B_0 c)/(Y-c) + B_0]$ and ψ is the standard normal density.*

The condition $a\, r_p < \pi/2$ does not involve q, because q enters only through the Gaussian factor.

Proof. Part (i). Fix a real y and b with $|b| < a$, and write $z = y + ibe_j$. The k-th term of $S(z)$ is $w_k e^{u_k + i\theta_k}$ with $u_k = \langle y, x_k\rangle - |x_k|^2/2$ real and $\theta_k = b\, x_{k,j}$. Since $|x_{k,j}| \le r_p$ we have $|\theta_k| \le a\, r_p < \pi/2$. Hence $\operatorname{Re} S(z) = \sum_k w_k e^{u_k}\cos\theta_k \ge \cos(a r_p)\, S_0$ with $S_0 = S(y) > 0$, so $\operatorname{Re} S > 0$. Next $|S(z)| \le S_0$ and $|\operatorname{Im} S(z)| \le \sin(a r_p) S_0$, so $|\arg S(z)| \le a r_p$ and $|\ln|S(z)|| \le |\ln S_0| + |\ln\cos(a r_p)|$. Each factor $\exp(\langle y, x_k\rangle - |x_k|^2/2)$ lies between $e^{-r_p|y| - r_p^2/2}$ and $e^{r_p|y|}$, and $|y| \le |y_1| + |y_2|$. So $|\ln S_0| \le |\ln s| + r_p^2/2 + r_p(|y_1| + |y_2|)$. Finally $|\ln s| \le 2|s-1|$ for $s \ge 1/4$: for $s \ge 1$ this is $\ln s \le s - 1$, and for $1/4 \le s < 1$ the function $k(s) = 2(1-s) + \ln s$ is concave with $k(1) = 0$ and $k(1/4) = 0.1137 > 0$, so $k \ge 0$ on the interval. Adding the pieces gives (i).

A one-variable lemma. Let f be holomorphic in $|\operatorname{Im} z| < a$, with $|f(x)| \le C(1+|x|)^{-2}$, with $f(x+ib) \to 0$ as $|x| \to \infty$ uniformly in $|b| \le a'$ for every $a' < a$, and with $M = \sup_{|b|<a} \int |f(x+ib)|\, dx < \infty$. Then

$$\Big| h \sum_m f(mh) - \int f \Big| \le \frac{2M}{e^{2\pi a/h} - 1}. \tag{10}$$

To see this let $\hat f(\xi) = \int f(x) e^{-i\xi x} dx$. For $\xi > 0$ and $0 < a' < a$ shift the contour to $\operatorname{Im} z = -a'$. The vertical sides vanish by the uniform decay, and $|e^{-i\xi(x - ia')}| = e^{-\xi a'}$. So $|\hat f(\xi)| \le M e^{-a'|\xi|}$, and the same holds for $\xi < 0$ by shifting upward. Let $a' \to a$. Put $g(t) = f(ht)$. Then $|g(t)| \le C(1 + h|t|)^{-2}$, and its transform in the convention of Stein and Weiss [40] is $\hat g(k) = \hat f(2\pi k/h)/h$ with $|\hat g(k)| \le (M/h) e^{-2\pi a'|k|/h}$. Both decay rates exceed 1, so Poisson summation holds: $\sum_m g(m) = \sum_k \hat g(k)$, that is $h\sum_m f(mh) = \sum_k \hat f(2\pi k/h)$. Removing $k = 0$, which equals $\int f$, and summing the geometric series gives (10).

Three Gaussian sums. For real c and $0 < h \le 1$, with $\psi = \varphi|_{\mathbb{R}}$:

$$h \sum_m \psi(mh - c) \le 1 + \eta, \tag{11}$$

$$h \sum_m |mh - c|\, \psi(mh - c) \le \sqrt{2/\pi} + 2h\psi(1) \le 0.7979 + 0.484\, h, \tag{12}$$

and, for integers $N \ge 1$ with $Y = Nh$, real c' and $c \ge |c'|$ with $Y - c \ge 1$, and $\alpha, \beta \ge 0$,

$$h \sum_{|mh|>Y} \psi(mh - c') \le \frac{2\psi(Y-c)}{Y-c},$$
$$h \sum_{|mh|>Y} \psi(mh - c')(\alpha + \beta|mh|) \le 2\psi(Y-c)\Big[\frac{\alpha + \beta c}{Y-c} + \beta\Big]. \tag{13}$$

For (11), Poisson summation gives $h\sum_m \psi(mh - c) = 1 + 2\sum_{k\ge1} \cos(2\pi k c/h)\, x^{k^2}$ with $x = e^{-2\pi^2/h^2} < 3\times10^{-9}$, and $2\sum_{k\ge1} x^{k^2} \le 2x(1 + x^3/(1-x)) \le 3x$. For (12), let $g(u) = |u|\psi(u)$. On $[0,\infty)$ it increases on $[0,1]$ and decreases on $[1,\infty)$, with maximum $\psi(1)$. Split the nodes $u \ge 0$ of the progression $mh - c$ into three classes: (1) $u + h \le 1$, (2) $u - h \ge 1$, (3) $1 - h < u < 1 + h$. A node of class (1) satisfies $g(u) \le h^{-1}\int_u^{u+h} g$ because g increases there, and a node of class (2) satisfies $g(u) \le h^{-1}\int_{u-h}^{u} g$. The cells used are disjoint. Class (3) has one or two nodes. A single node contributes at most $h\psi(1)$. Two nodes $u_p < 1 < u_{p+1}$ span a cell that classes (1) and (2) do not use, on which g is at least the smaller of its end values. So $h\,(g(u_p) + g(u_{p+1})) \le \int_{u_p}^{u_{p+1}} g + h\psi(1)$. In every case the nodes $u \ge 0$ contribute at most $\int_0^\infty g + h\psi(1)$, and the nodes $u \le 0$ likewise. Since $\int |u|\psi = \sqrt{2/\pi} = 0.79788$ and $2\psi(1) = 0.48394$, we get (12). For (13), the function $\psi(u)(\alpha' + \beta u)$ with $\alpha' = \alpha + \beta c$ decreases on $[1,\infty)$, because its derivative is $\psi(u)[\beta - u(\alpha' + \beta u)] \le 0$ there. The cell to the left of each tail node lies in $[Y - c, \infty)$, so the tail sum is at most $\int_{Y-c}^\infty \psi(u)(\alpha' + \beta u)\, du = \alpha'(1 - \Phi(Y-c)) + \beta\psi(Y-c) \le \psi(Y-c)\,[\alpha'/(Y-c) + \beta]$ by the Mills bound $1 - \Phi(t) \le \psi(t)/t$. Two tails give the factor 2.

Part (ii). Split $I - T_h F = (I_1 - T_1)(I_2 F) + T_1(I_2 - T_2)F$, where I_j and T_j are the integral and the infinite trapezoid sum in y_j. All sums and integrals converge absolutely. For $z = y_1 + ib$ with $|b| < a$ and real y_2 we have $|\varphi(z - q_1)| = \psi(y_1 - \operatorname{Re} q_1) e^{(b - \operatorname{Im} q_1)^2/2}$, so by part (i)

$$|F(y_1 + ib, y_2)| \le E\, \psi(y_1 - \operatorname{Re} q_1)\, \psi(y_2 - \operatorname{Re} q_2) \times [A_0 + B_0(|y_1| + |y_2|)], \tag{14}$$

and the same with the roles of the coordinates exchanged. For the first term, $G(z) = \int F(z, y_2)\, dy_2$ is holomorphic (dominated convergence and Morera's theorem), and by (14) and $\mathbb{E}|c + Z| \le |c| + \sqrt{2/\pi}$ we get $M_1 := \sup_b \iint |F(x + ib, y_2)|\, dy_2\, dx \le E[A_0 + B_0(Q_1 + Q_2 + 1.6)]$; the same bound controls $|G(x + ib)|$ uniformly in b, which gives the decay hypotheses. The one-variable lemma gives $|(I_1 - T_1)G| \le 2M_1/(e^{2\pi a/h} - 1)$. For the second term apply it to $z \mapsto F(mh, z)$ at each grid point, with $M_2(y_1) \le E\, \psi(y_1 - \operatorname{Re} q_1)[A_0 + B_0|y_1| + B_0(Q_2 + \sqrt{2/\pi})]$. Summing over m with (11), (12) and $|mh| \le Q_1 + |mh - \operatorname{Re} q_1|$ gives $h \sum_m M_2(mh) \le E(1+\eta)[A_0 + B_0(Q_1 + Q_2 + 1.6 + 0.484h)]$. Adding the two terms and using $1 + (1 + \eta) \le 2(1 + \eta)$ gives (ii).

Part (iii). For real y we have $|F(y)| \le e^{\sigma^2} \psi(y_1 - \operatorname{Re} q_1)\psi(y_2 - \operatorname{Re} q_2)[A_0 + B_0(|y_1| + |y_2|)]$. The grid points outside the square lie in $\{|mh| > Y\} \cup \{|nh| > Y\}$. For the first set, the sum is at most e^{σ^2} times $\big[h \sum_{|mh|>Y} \psi(\cdot)(A_0 + B_0|mh|)\big]\big[h \sum_n \psi(\cdot)\big] + B_0\big[h \sum_{|mh|>Y} \psi(\cdot)\big]\big[h \sum_n |nh|\psi(\cdot)\big]$. Bounds (13), (11) and (12) give $(1+\eta)\tau_1(Q_1) + B_0\tau_0(Q_1)((1+\eta)Q_2 + 0.7979 + 0.484h)$. The second set gives the same with the indices exchanged. □

Let $m_{\min}$ be the smallest of the masses m_j. The same bounds hold for the derivative of $\log S$ in a mass, with $(A_0, B_0) = (1/(m_{\min} \cos ar_p), 0)$, and in an angle, with $(A_0, B_0) = (r_p(a + r_p)/\cos ar_p,\ r_p/\cos ar_p)$; the quadrature error of derivatives along the curve follows by Cauchy estimates on a disc of radius 0.5. At real points the bound is evaluated with $\sigma = 0$, $Q_1 = r_p$ and $Q_2 = r_m$. For the existence step: $h = 1/28$, $N = 658$ ($Y = 23.5$), $a = 1.45$, 576-bit precision, total bound on D_π of 1.6×10^{-109}. For the curve step: $h = 1/16$, $Y = 18$, 256 bits, the same $a = 1.45$, bounds 4.7×10^{-62} on D_π and 1.4×10^{-60} on $\partial_t^2 D_\pi$. The Taylor remainders along the curve use Cauchy discs of radius 1 in t, on which $|\operatorname{Re} q_j| \le r_p \cosh 1$ and $|\operatorname{Im} q_j| \le r_p \sinh 1$; the bound on $|D_\pi - C|$ there comes from the same trapezoid sums with $|\varphi_2(y - q)| = \varphi_2(y - \operatorname{Re} q) e^{|\operatorname{Im} q|^2/2}$.

Existence. The starting point has residual 1.3×10^{-50} in the exact equations; three Newton steps on midpoints (not part of the proof) give $\hat v$ with rigorous residual $|F(\hat v)| \le 1.96 \times 10^{-108}$. With Y_0 the inverse of the midpoint of the interval Jacobian $J(B)$ over $B = \hat v \pm r$, $r_i = \max(10|Y_0 F(\hat v)|_i, 10^{-7} \max_k |Y_0 F(\hat v)|_k)$, the Krawczyk operator $K(B) = \hat v - Y_0 F(\hat v) + (I - Y_0 J(B))(B - \hat v)$ satisfies $\|I - Y_0 J(B)\|_\infty \le 8.97 \times 10^{-22}$ and $\max_i |K(B) - \hat v|_i / r_i \le 0.100$, so $K(B) \subset \operatorname{int} B$ and B contains exactly one solution.

Boundary. On each of 32 intervals of half-width $\pi/128$ the function $t \mapsto D_\pi(z(t)) - C$ is enclosed, uniformly for $v \in B$, by its Taylor polynomial of order 36, a Cauchy remainder on a disc of radius 1, and the quadrature bound. The remainder is 1.07×10^{-58} for the function and 2.37×10^{-52} for its second derivative. Zones $|t - \hat t_j| \le r_j + 0.004$ around the atoms and the Taylor breakpoints give 48 base intervals, 38 free and 10 in atom zones, subdivided adaptively to 230 pieces. The last interval is closed at the ball enclosing $\pi/2$. For every $v \in B$ we have $D_\pi - C \in [-1.12760361458, -1.12760361456] \times 10^{-50}$ at $(r_p, 0)$.

Interior. The conditional covariance satisfies $\operatorname{Cov}(X \mid y) \preceq \max_k |x_k|^2 I$ and $\max_k |x_k|^2 = \cos^2 t_1 + r_m^2 \sin^2 t_1 = 0.99984$, so $\nabla^2 D_\pi \succ 0$; every interior point is a strict convex combination of two boundary points, so $D_\pi < \max_{\partial E} D_\pi \le C$ there.

ACKNOWLEDGMENT

The author thanks Sarah Marzen for helpful comments.

Use of AI. Claude (Anthropic) was used as an assisting tool during the preparation of this manuscript, primarily for computational support and language editing. The research questions, mathematical ideas, methodology, interpretation of results, and overall direction of the work were developed by the author. All computational results and certificates reported in the manuscript were recomputed and verified by the author, and the cited works were checked by the author. The author is responsible for the content and conclusions of this work.

REFERENCES

[1] K. Rose, "Deterministic annealing for clustering, compression, classification, regression, and related optimization problems," *Proc. IEEE*, vol. 86, no. 11, pp. 2210–2239, 1998.

[2] J. G. Smith, "The information capacity of amplitude- and variance-constrained scalar Gaussian channels," *Inf. Control*, vol. 18, no. 3, pp. 203–219, 1971.

[3] M. N. Ghosh, "Uniform approximation of minimax point estimates," *Ann. Math. Statist.*, vol. 35, pp. 1031–1047, 1964.

[4] G. Casella and W. E. Strawderman, "Estimating a bounded normal mean," *Ann. Statist.*, vol. 9, no. 4, pp. 870–878, 1981.

[5] I. M. Johnstone, *Gaussian Estimation: Sequence and Multiresolution Models*, draft, 2011.

[6] Y. Polyanskiy and Y. Wu, "Self-regularizing property of nonparametric maximum likelihood estimator in mixture models," arXiv:2008.08244, 2020.

[7] Y. Polyanskiy and M. Sellke, "Nonparametric MLE for Gaussian location mixtures: certified computation and generic behavior," arXiv:2503.20193, 2025.

[8] S. Shamai (Shitz) and I. Bar-David, "The capacity of average and peak-power-limited quadrature Gaussian channels," *IEEE Trans. Inf. Theory*, vol. 41, no. 4, pp. 1060–1071, 1995.

[9] A. Dytso, M. Al, H. V. Poor, and S. Shamai (Shitz), "On the capacity of the peak power constrained vector Gaussian channel: An estimation theoretic perspective," *IEEE Trans. Inf. Theory*, vol. 65, no. 6, pp. 3907–3921, 2019.

[10] B. Rassouli and B. Clerckx, "On the capacity of vector Gaussian channels with bounded inputs," *IEEE Trans. Inf. Theory*, vol. 62, no. 12, pp. 6884–6903, 2016.

[11] H. Wang, L. Barletta, and A. Dytso, "An improved lower bound on cardinality of support of the amplitude-constrained AWGN channel," *IEEE Trans. Inf. Theory*, vol. 72, no. 8, pp. 5392–5399, 2026. Preprint: arXiv:2512.22691.

[12] A. Dytso, S. Yagli, H. V. Poor, and S. Shamai (Shitz), "The capacity achieving distribution for the amplitude constrained additive Gaussian channel: An upper bound on the number of mass points," *IEEE Trans. Inf. Theory*, vol. 66, no. 4, pp. 2006–2022, 2020.

[13] H. H. Mattingly, M. K. Transtrum, M. C. Abbott, and B. B. Machta, "Maximizing the information learned from finite data selects a simple model," *Proc. Natl. Acad. Sci. USA*, vol. 115, no. 8, pp. 1760–1765, 2018.

[14] M. C. Abbott and B. B. Machta, "A scaling law from discrete to continuous solutions of channel capacity problems in the low-noise limit," *J. Stat. Phys.*, vol. 176, no. 1, pp. 214–227, 2019.

[15] H. Wang, "Super-linear growth of the capacity-achieving input support for the amplitude-constrained AWGN channel," arXiv:2510.20723, 2025.

[16] L. Barletta and A. Dytso, "Support size of ϵ-capacity-achieving inputs for the amplitude-constrained AWGN channel," arXiv:2604.14915, 2026.

[17] L. Barletta and A. Dytso, "On the evolution of the capacity-achieving input support for the amplitude-constrained AWGN channel," arXiv:2609.10039, 2026.

[18] A. Dytso, L. Barletta, and G. Kramer, "On 2×2 MIMO Gaussian channels with a small discrete-time peak-power constraint," in *Proc. IEEE Int. Symp. Inf. Theory (ISIT)*, 2024, pp. 2365–2370, doi: 10.1109/ISIT57864.2024.10619699. Extended version with appendices: arXiv:2401.17084, 2024.

[19] A. Dytso, M. Goldenbaum, H. V. Poor, and S. Shamai (Shitz), "Amplitude constrained MIMO channels: properties of optimal input distributions and bounds on the capacity," *Entropy*, vol. 21, no. 2, Art. no. 200, 2019, doi: 10.3390/e21020200.

[20] J. Kiefer and J. Wolfowitz, "Consistency of the maximum likelihood estimator in the presence of infinitely many incidental parameters," *Ann. Math. Statist.*, vol. 27, no. 4, pp. 887–906, 1956.

[21] T. H. Chan, S. Hranilovic, and F. R. Kschischang, "Capacity-achieving probability measure for conditionally Gaussian channels with bounded inputs," *IEEE Trans. Inf. Theory*, vol. 51, no. 6, pp. 2073–2088, 2005.

[22] B. G. Lindsay, "The geometry of mixture likelihoods: a general theory," *Ann. Statist.*, vol. 11, no. 1, pp. 86–94, 1983.

[23] H. Robbins, "An empirical Bayes approach to statistics," in *Proc. Third Berkeley Symp. Math. Statist. Probab.*, vol. 1, 1956, pp. 157–163.

[24] L. D. Brown, "Admissible estimators, recurrent diffusions, and insoluble boundary value problems," *Ann. Math. Statist.*, vol. 42, pp. 855–903, 1971.

[25] C. Hatsell and L. Nolte, "Some geometric properties of the likelihood ratio," *IEEE Trans. Inf. Theory*, vol. 17, no. 5, pp. 616–618, 1971.

[26] A. Thangaraj, G. Kramer, and G. Böcherer, "Capacity bounds for discrete-time, amplitude-constrained, additive white Gaussian noise channels," *IEEE Trans. Inf. Theory*, vol. 63, no. 7, pp. 4172–4182, 2017.

[27] P. J. Bickel, "Minimax estimation of the mean of a normal distribution when the parameter space is restricted," *Ann. Statist.*, vol. 9, no. 6, pp. 1301–1309, 1981.

[28] N. Sharma and S. Shamai (Shitz), "Transition points in the capacity-achieving distribution for the peak-power limited AWGN and free-space optical intensity channels," *Probl. Inf. Transm.*, vol. 46, no. 4, pp. 283–299, 2010.

[29] S. Karlin, *Total Positivity, Vol. I*. Stanford, CA, USA: Stanford University Press, 1968.

[30] M. Gasca and J. M. Peña, "Total positivity and Neville elimination," *Linear Algebra Appl.*, vol. 165, pp. 25–44, 1992.

[31] W. B. Jones, O. Njåstad, and W. J. Thron, "Moment theory, orthogonal polynomials, quadrature, and continued fractions associated with the unit circle," *Bull. London Math. Soc.*, vol. 21, no. 2, pp. 113–152, 1989.

[32] B. Simon, *Orthogonal Polynomials on the Unit Circle, Part 1: Classical Theory*. Providence, RI, USA: Amer. Math. Soc., 2005.

[33] F. Johansson, "Arb: efficient arbitrary-precision midpoint-radius interval arithmetic," *IEEE Trans. Comput.*, vol. 66, no. 8, pp. 1281–1292, 2017.

[34] L. N. Trefethen and J. A. C. Weideman, "The exponentially convergent trapezoidal rule," *SIAM Review*, vol. 56, no. 3, pp. 385–458, 2014.

[35] R. Krawczyk, "Newton-Algorithmen zur Bestimmung von Nullstellen mit Fehlerschranken," *Computing*, vol. 4, pp. 187–201, 1969.

[36] R. E. Moore, "A test for existence of solutions to nonlinear systems," *SIAM J. Numer. Anal.*, vol. 14, no. 4, pp. 611–615, 1977.

[37] S. M. Rump, "Verification methods: Rigorous results using floating-point arithmetic," *Acta Numerica*, vol. 19, pp. 287–449, 2010.

[38] J. C. Berry, "Minimax estimation of a bounded normal mean vector," *J. Multivariate Anal.*, vol. 35, no. 1, pp. 130–139, 1990.

[39] I. J. Schoenberg, "On Pólya frequency functions," *J. Analyse Math.*, vol. 1, pp. 331–374, 1951.

[40] E. M. Stein and G. Weiss, *Introduction to Fourier Analysis on Euclidean Spaces*. Princeton, NJ, USA: Princeton University Press, 1971.